

Title: Decades matter: Agricultural diversification increases financial profitability, biodiversity, and ecosystem services over time.

Authors: Estelle Raveloaritiana^{1,2,*}, Thomas Cherico Wanger^{1,2,3,4,5*}

¹ Sustainable Agricultural Systems & Engineering Laboratory, School of Engineering, Westlake University, Hangzhou, China.

² Key Laboratory of Coastal Environment and Resources of Zhejiang Province, Westlake University, Hangzhou, China.

³ Faculty of Science, Technology & Medicine, University of Luxembourg, Luxembourg

⁴ Global Agroforestry Network, Hangzhou, China

⁵ China Rice Network, Hangzhou, China

*Corresponding authors: eraveloaritiana@gmail.com (ER); tomcwanger@gmail.com (TCW)

Abstract (194/200 words maximum)

Sustainable agriculture in the 21st century requires the production of sufficient food while reducing the environmental impact and safeguarding human livelihoods ^{1,2}. Many studies have confirmed agricultural diversification practices such as intercropping, organic farming and soil inoculations as a suitable pathway to achieve these goals ^{3,4} but long-term viability of socioeconomic and ecological benefits is uncertain ^{5,6}. Here, we quantified the long-term effects of agricultural diversification practices on socioeconomic and ecological benefits based on 50 years of data from 184 meta-analyses and 4,260 effect sizes. We showed that, with neutral crop yield over time, financial profitability, most variables related to biological communities, all aspects of soil quality, and carbon sequestration benefits increased by up to 2823% over 20 years of practice. Non-crop diversification practices and the use of organic amendments increased benefits by up to 2000% after 50 years. A trade-off analysis between yield and other services showed win-win outcomes during the first 25 years. Our synthesis provides the urgently needed evidence for farmers and other decision-makers that diversification increases long-term profitability, biodiversity, and climate mitigation benefits, and therefore, allows upscaling diversification for climate change mitigation and global food system transformation ⁷.

Keywords: diversified farming, ecosystem services, biodiversity, yield, financial profitability, trade-off over time, practices duration

Main

Transforming the global food system requires nature-positive production systems^{4,8} based on agricultural diversification to achieve sustainable food production^{9–11}. Agricultural diversification is based on farming practices, such as crop rotation, agroforestry, or organic farming, that purposely integrate functional diversity into production systems to enhance or maintain the services essential to crop production³. Several syntheses have confirmed the socioeconomic (i.e., yield and profitability) and environmental (i.e., biological communities, soil quality, and climate mitigation) benefits of agricultural diversification practices for crop production worldwide^{5,6}. In Asia, for example, rice crop diversification practices reduce pesticide use and increase biocontrol and crop yields¹². These benefits have incentivized stakeholder groups to implement agricultural diversification across different countries. In China, agricultural diversification was recently promoted into the country's major agricultural policies¹³. In Europe and the USA, several agricultural diversification practices, such as organic farming and crop/non-crop diversification, are advocated in agricultural policies for biodiversity conservation and climate change mitigation^{14–16}. Despite these successes, the long-term effects of agricultural diversification on socioeconomic and environmental factors remain unquantified and impede large-scale implementation by risk-averse farmers and promotion through policy makers.

Stakeholder groups require long-term evidence that diversified agricultural production can maintain and ideally improve livelihoods, economy, and the environment to mitigate climate change and external stressors^{17–19}. For instance, in staple foods like rice, the socioeconomic and environmental benefits of diversified rice farming are now well quantified²⁰, but implementation success will depend on understanding temporal dynamics of such benefits¹⁷. In cash crops like cocoa, smallholder farmers have made long-term investments into their tree plantations and may be reluctant to transition towards diversified agroforestry systems without understanding long-term effects on profitability and yields²¹. More generally, global food insecurity requires policy decisions that assure reaching the Sustainable Development Goals now more than ever^{7,22,23}. Moving beyond non-temporal analyses and quantifying the long-term effects of agricultural diversification practices on socioeconomic and environmental factors - and

the trade-offs between them is, hence, decisive to the successful implementation of a global food system transformation ^{22,24,25}.

Here, we conducted a global synthesis of meta-analyses to quantify the effects of agricultural diversification practices on socioeconomic and environmental factors for up to 50 years. Based on data from 184 meta-analyses with 4,260 effect sizes based on 17,989 original studies (Supplementary Figure 1), we quantified the long-term effects on socioeconomic factors (crop yield and financial profitability), biological communities (biodiversity, pollination, and pest control), soil quality (water regulation, soil fertility, and nutrient cycling), and climate change mitigation (carbon sequestration and climate regulation). We also analysed the temporal trade-offs between crop yield and ecosystem services (Methods).

Socioeconomic, biological community, soil quality, and climate change mitigation benefits mostly increase over time

Our second-order meta-regression models showed that agricultural diversification practices significantly increased benefits for seven out of ten socioeconomic and environmental factors in the long term (Fig. 1, Extended Data Table 1). The analysis of socioeconomic factors showed that while agricultural diversification did not reduce crop yield over time (Fig. 1), financial profitability increased significantly by 172% over 20 years compared to short-term effects (< 3 years, Extended Data Fig. 2, Extended Data Table 2). These findings show that the crop yield of diversified farming systems stabilizes over time ^{26,27}, while farmers' income improves and, hence, highlighting the socio-economic viability of agricultural diversification in the long run ⁹.

For biological communities, agricultural diversification significantly increased biodiversity and pollination benefits by 231% and 2823%, respectively, over 20 years (Fig. 1, Extended Data Table 2). These patterns provide evidence that agricultural diversification can play an important role not only in bending the curve of biodiversity losses from agriculture ²⁸ but also in improving pollination services ²⁹, which is highly important for global food security ³⁰. However, the effects of diversification on pest control decreased over time (Fig. 1, Extended Data Table 1-2), but with an increased probability of neutral effects over the years (Extended Data Fig. 1). Diversified farming

may, hence, provide concomitant benefits to pest control agents and pests ³¹, which leads to overall negative trends.

All effect sizes of soil quality and climate change mitigation, except for climate regulation, were significantly positive but did not significantly change over time (Fig.1). These patterns suggest that benefits of agricultural diversification for water regulation, soil fertility, nutrient cycling, and carbon sequestration are maintained in the long run. The transformation from simple monocultures to diversified farming systems, therefore, can improve soil quality and carbon stocks in agricultural lands sustainably ^{32,33}.

Overall, our results provide evidence that agricultural diversification can help to achieve sustainable and climate resilient food production ^{4,34} with financial profitability, biodiversity and pollination being 100%, 56% and 1164%, respectively, higher after 20 years compared to non-temporal analysis results (Extended Figure 2, Extended Table 2). Our analyses show that non-temporal analyses vastly underestimate the full potential that agricultural diversification can bring for socioeconomic and environmental factors after just two decades ^{22,34}.

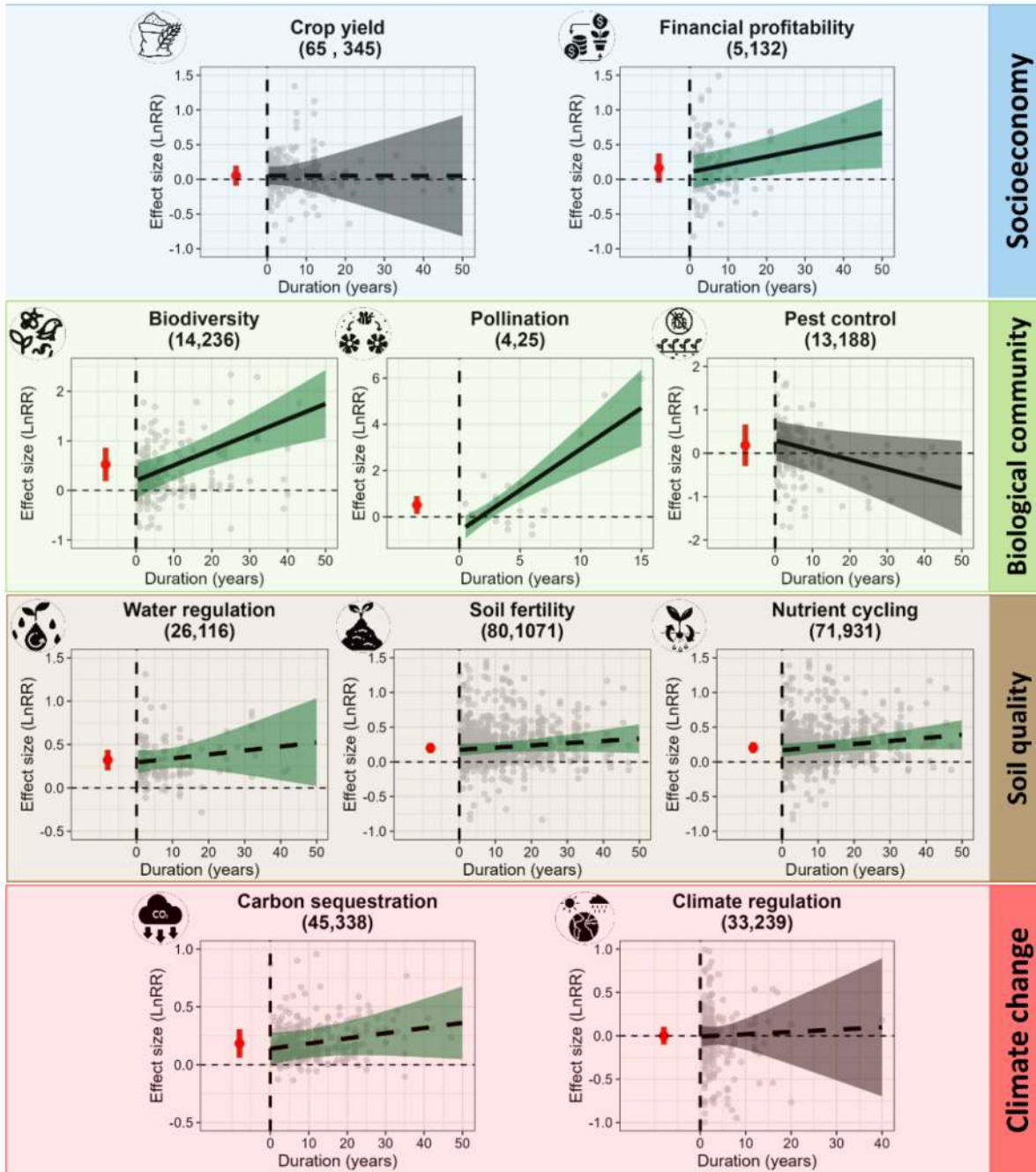


Fig. 1| Effects of agricultural diversification on socioeconomic, biological community, soil quality, and climate change-related variables over time compared to non-temporal effects. Red dot and bars represent non-temporal means with 95% confidence intervals (CI), respectively, *sensu* Tamburini et al. 2020. Respective solid or dashed lines inside ribbons represent significant and non-significant change of effect sizes (LnRR) over time; ribbons are the 95% CI. Ribbon colours: grey = effects of diversification practices are neutral; green = significant positive effect of diversification. In brackets are the number of studies and effect sizes.

Our meta-regression analyses on the effects of six diversification practices on socioeconomic factors, biological community, soil quality, and climate change categories showed that almost half of the response variables became significantly positive or increased over time (dark green in Fig. 2, Extended Data Table 3-4). Non-crop diversification and organic amendments increased response variables in all four categories over time (Fig. 2, Extended Data Table 4), which provides realistic opportunities to address the twin challenges of biodiversity loss^{28,35} and climate mitigation^{36,37}. In addition, the long-term maintenance or significant improvement of soil fertility and nutrient cycling shows enhancement of soil quality^{36,38,39}. Long-term benefits of organic amendments highlight the potential of this practice to reduce chemical inputs, and hence, pollution and health repercussions^{40,41}. Therefore, these practices constitute the best options for shifting away from unsustainable and towards more biodiversity and nature-friendly farming practices^{9,22}.

Among all the types of agricultural diversification practices, crop diversification was the only practice that significantly increased financial profitability and resulted in neutral yields over time (Fig. 2, Extended Data Table 4). Crop diversification can, hence, improve farmers' income over time⁴² and ensure the long-term stability of crop yield²⁷. While pest control on crop diversification decreased significantly with time, there was no significant difference compared to conventional agriculture for up to 50 years (Fig. 1, Extended Data Table 4). This result indicates that some known benefits for instance of crop rotation on pest control may be counteracted⁴³, where environmental complexity increases pest incidence and herbivory richness⁴³⁻⁴⁵. Reduced tillage showed slight yield increases over time but significant long-term benefits for carbon sequestration, while organic farming did not provide significant long-term benefits (Fig. 2, Extended Data Table 4). Overall, mainstreaming agricultural diversification and the combination of various practices may be a very effective and long-term strategy for a global food systems transformation¹¹. For instance, non-crop diversification strategies seem to hold most benefits as an individual practice, while organic farming shows an incremental increase in biodiversity benefits over time, but has consistently lower yields (but see⁴⁶). An effective combination of practices may help to transform some of the neutral into positive long-term benefits⁶.



Fig. 2| Effects of individual diversification practices on socioeconomic, biological community, soil quality, and climate change-related variables over time compared to non-temporal effects. Red dot and bars represent means from non-temporal analyses with 95% confidence intervals (CI), respectively, *sensu* Tamburini et al. 2020. Respective solid or dashed lines inside ribbons represent significant and non-significant change of effect sizes (LnRR) over time; ribbons are the 95% CI. Ribbon colours: grey = effects are neutral over time; green = significant positive effect of diversification. For non-crop diversification, financial profitability and climate regulation predictions cover three years and one year, respectively. For organic farming, the water regulation predictions cover 33 years. Empty panels indicate research gaps due to data unavailability.

Long term trade-offs between crop yield and other ecosystem services

We predicted trade-offs between crop yield and other services over time based on hierarchical regression models, in which all effect sizes were divided into loss and win outcomes (Methods). Overall, we detected a win-win situation between crop yield and all other services, with a probability higher than 60% within the first 25 years of agricultural diversification practices, and then a lose yield-win ecosystem services was the most likely outcome (Fig. 3A, Extended Data Table 5).

When examining the trade-off between yield and individual services, we found that increasing win-win situations with biodiversity, water regulation, and climate regulation were predominant (Fig. 3B-J, Extended Data Table 5). These results show that the positive and desirable outcomes of agricultural diversification practices not only remain but also become more probable over time ⁸. Our analyses also showed win-win situations with financial profitability, soil fertility, nutrient cycling, and carbon sequestration for 5, 24, 25 and 15 years, respectively (Fig. 3) and lose yield-win ecosystem service situations becoming more common afterwards. Our results indicate that compared to monocultures and conventional agriculture, agricultural diversification can improve crop yield while increasing farmers' income, soil nutrients, and capturing and storing carbon from the atmosphere, thus achieving nature-positive systems and equitable livelihoods ^{4,47,48}. Government agri-environmental schemes should, hence, compensate for lower yields in diversified systems ⁴⁹.

We also found a predominance of lose yield-win pest control and pollination relationships over the first ten years (Fig. 3), indicating that increases in pollination and pest control services do not always translate into higher crop yields. It is noteworthy, however, that while there is not necessarily a yield increase with pollination, 70% of all crops globally depend on natural pollination; hence, pollination is critical for yield stability ^{30,50}. In addition, some studies have shown that pollination and pest control services can increase yield quality, thus improving the market value of crop yield ⁵¹. Our analysis also revealed a temporal increase of the lose-win relationship between crop yield and carbon sequestration. Thus, climate financing for diversified agriculture will account for such

carbon-yield trade-offs and requires even more advocacy for diversification and agroecology at climate summit discussions.

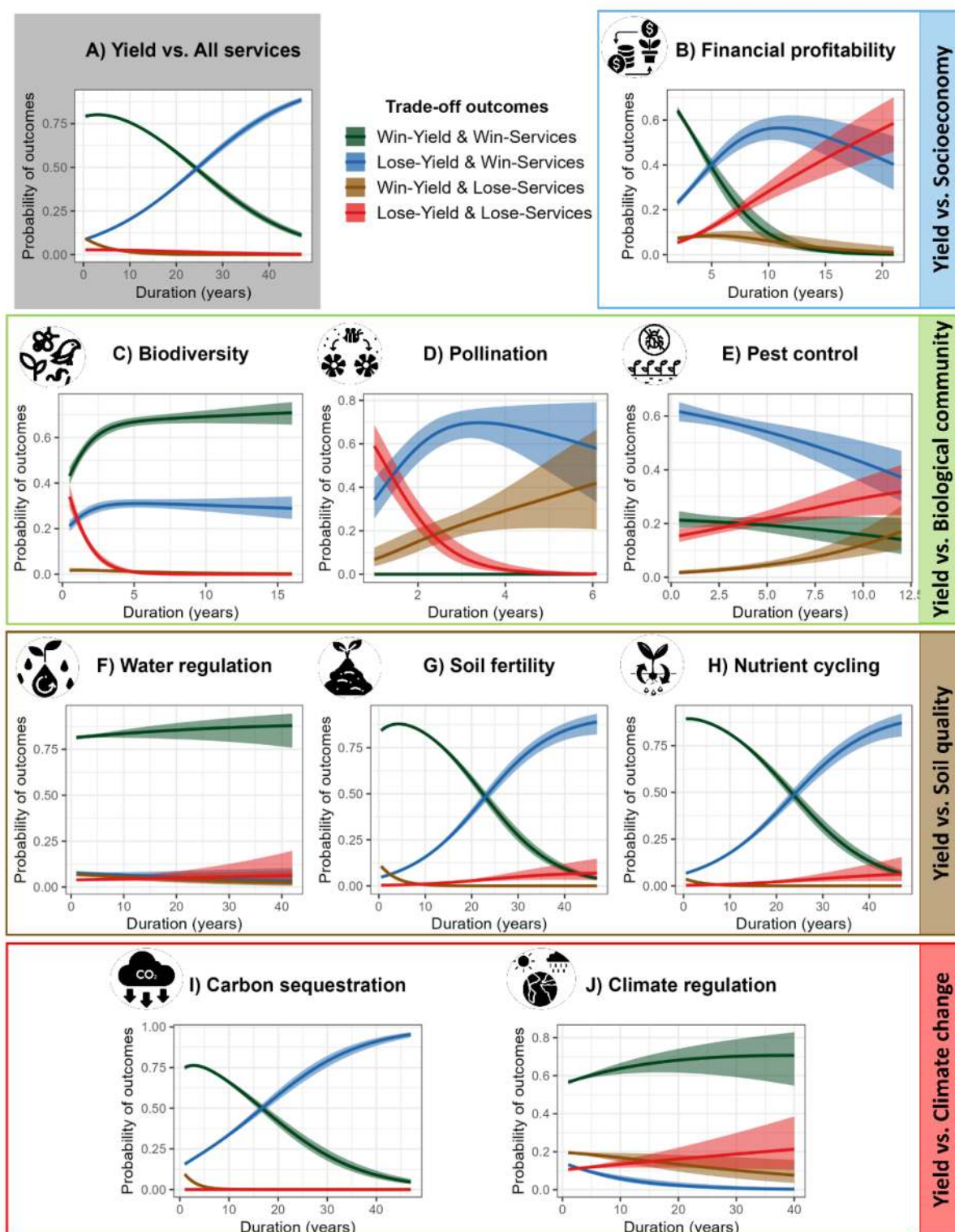


Fig. 3| Trade-offs for agricultural diversification implementation over time: crop yield vs.

overall (A) and individual ecosystem services in the four categories of socioeconomy (B), biological community (C-E), soil quality (F-H), and climate change (I-J).

Research gaps

Our work quantifies diversification effects for up to 50 years and allows us to identify temporal research gaps for the first time. First, short-term meta-analyses (0-10 years) on diversification effects on socioeconomic, biological community, soil quality, and climate change related variables were almost twice more common (137 studies and 2479 effect sizes) than long-term (>10 years) meta-analyses (74 studies and 1140 effect sizes). Likely, primary studies with long-term data are lacking, as funding opportunities for studies or experiments of >10 years are limited⁵², but critically important to understand, for instance, diversification effects on pollination, climate regulation, and financial profitability (see Fig. 1 and Extended Data Fig. 3). Second, our trade-off analysis contained less data (43 studies, 1448 effect sizes) than the general meta-regressions (148 studies, 3619 effect sizes, Extended data Fig. 3-4). More studies are needed that concomitantly focus on diversification effects on yield or other services. Lastly, for individual diversification practices, we found no data on the effects of organic amendments and reduced tillage on financial profitability, pollination, and pest control over time (Fig. 2). Nevertheless, the use of these two practices is highly prevalent and advocated globally^{34,38}. There were also no data on organic farming effects on climate regulation and almost no effect sizes for long-term inoculation effects (see Fig. 2). Overall, syntheses based on long-term studies require innovative funding mechanisms from various sources for ultimately managing all trade-offs in diversified farming systems to maintain food security, alleviate poverty, and reduce environmental impacts^{53,54}.

A long-term perspective to implement agricultural diversification.

We show that non-temporal analyses have underestimated the benefits of agricultural diversification by up to 1164%, particularly for financial profitability, biodiversity, and pollination. Our work is the first quantification of the long-term benefits of agricultural diversification that can inform researchers, policymakers, and practitioners on urgently needed implementation strategies⁸.

For the research community, it is important to focus on long-term projects that can disentangle the complex relationships between yield and various ecosystem services linked to biodiversity loss and climate change ^{9,17}. Intuitively, a long-term focus will also address many remaining short-term research gaps on the way ⁵. Clearly, addressing trade-offs between different ecosystem services requires multidisciplinary collaborations and international partnerships such as the Agroecology Coalition, which can play a key role towards rapidly advancing progress of the Sustainable Development Goals ^{22,55}. Naturally, addressing long-term and applied research questions will require long-term funding mechanisms from the public and private sectors ⁵².

For policymakers, our results provide critical evidence of how agricultural diversification can provide long-term support for biodiversity conservation, climate change mitigation, and at least maintaining yields ³⁴. Biodiversity benefits increase over time without affecting crop yield, despite the commonly expected trade-off associated with biodiversity integration in farmlands ^{56–58}. The contribution of diversified farming systems to climate change mitigation increases over time, creating win-win solutions between crop yield and climate regulation. Awareness of agricultural diversification is increasing in Europe and North America, but efforts need to be enhanced, particularly in Asia and Africa, to achieve global-scale targets of climate change mitigation and sustainable food production ⁴⁹. Agri-environmental schemes and equivalent policies globally should compensate for higher yield gaps in diversified systems to alleviate trade-offs where they exist. Then policies can help restoring degraded and agricultural lands more effectively to reach climate and biodiversity targets ⁵⁹.

Our work enables farmers to understand short- and long-term benefits of agricultural diversification compared to conventional agriculture or monocultures. A core argument for implementation is a sustainable production strategy that can maintain crop yields and increase financial profitability over time and benefit the environment.. It also enables broader landscape finance strategies, as our results provide detailed evidence of the anticipated trade-offs of agricultural diversification implementation and the associated risks ⁵⁸. Agricultural diversification provides further opportunities to shift towards a truly integrated strategy ³⁵, whereby local culture and practices are accounted for, and ecosystem services are fully utilized for diverse livelihoods beyond crop-related

productions⁶⁰. In conclusion, our synthesis of the long-term benefits and trade-offs of agricultural diversification globally adds another urgently needed piece of the transformation puzzle for scientists, policymakers, and farmers to make well-informed and urgent decisions for sustainable agricultural systems.

Methods

Literature search and data compilation

We systematically searched for peer-reviewed meta-analysis papers analysing the effects of agricultural diversification on socioeconomic, biological community, soil quality and climate change, using Web of Science (<https://www.webofscience.com/>) and Scopus (<https://www.scopus.com/>). More specifically, we searched for all papers published until December 31, 2022, using keywords related to agricultural diversification practices and socioeconomic and ecological factors related to farming practices, and set meta-analysis as the type of study (for search strings, see Supplementary Table 1). Our research yielded 5,008 papers, which we then screened on Rayyan (<https://www.rayyan.ai/>) to determine which papers met the criteria for full-text screening. We considered a paper relevant for full-text screening if it was about agricultural production systems, potentially diversified farming systems, and focused on crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, or climate regulation, and if it was not a primary study. These criteria resulted in 3,692 papers being excluded, leaving 1,316 papers for full-text screening (Supplementary Figure 1).

We screened the full-text of 1,316 papers individually and searched for the following information as inclusion criteria: (1) the study should be a meta-analysis with at least one effect size comparing diversified farming systems to non-diversified practices. (2) The effect size should measure at least one of the following related variables: crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, and climate regulation (for more details on the variables, see Supplementary Table 2). (3) The number and list of original studies, as

well as the number of replicates for each effect size, should be reported. (4) The study should also include information on the duration of diversification practices. In the case of multiple response variables from the same study, we ensured that each response is associated with an independent factor without the possibility of redundancy from one response to another and causing pseudo-replication of the underlying data and effects. These criteria narrowed the list to 192 meta-analyses for data extraction.

We extracted the mean and variance of effect sizes expressed in log response ratio (lnRR), response ratio (RR), and percentage of change, and noted the qualitative effect (i.e., whether it was neutral, significantly positive, or significantly negative). We used Plot Digitizer (<http://plotdigitizer.sourceforge.net/>) to extract data from the figures. We converted the effect sizes from RR and percentage of change into lnRR to obtain a uniform measurement. When the effect size was presented in standardized mean difference (SMD) such as *Hedges' d* or other effect sizes that could not be converted into lnRR without original data, we only extracted the trend (negative or positive) of the effect size as well as the statistical significance, that is, negative, neutral, or positive.

For each effect size, we noted the duration (years) of agricultural diversification practices as well as the number of original comparisons. When the duration of practice was an interval, we considered the median duration (e.g. for 5-10 years, the duration would become 7.5 years). When the duration was indicated as ">x" we added two years, so >10 years would become 12 years to have a single duration for the corresponding effect size. We then categorised each response variable (effect size) into crop yield, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration and climate regulation (see Supplementary Table 2). We also categorized agricultural diversification practices into six categories: non-crop diversification, crop diversification, organic amendment, reduced tillage, inoculation, and organic farming (see Supplementary Table 3).

A strong synthesis should not repeat the original studies to guarantee that the effect sizes from all the studies are independent. We avoided including meta-analyses with similar underlying data by first running an independence analysis of 192 studies based on the references of their original studies. We calculated the percentage of original

references shared by papers and identified 24 papers with more than 30% of shared original studies. We examined the focus and duration of the effect sizes in these studies. When two or more papers had the same response variable within the same duration, we retained the effect sizes from the most recent study. Using these criteria, we excluded eight papers, resulting in 184 meta-analysis papers with 4,260 effect sizes from 17,989 studies and 169,555 comparisons (Supplementary Figure 1). We evaluated each study's quality based on the methods used and the reported results (for more details, see Supplementary Table 4).

Data analysis

We synthesized the effects of agricultural diversification on different categories of response variables in overall and per agricultural diversification practice. We used second-order meta-analysis based on hierarchical meta-regression models, weighted with number of comparisons behind each effect size to account for the number of observations behind each effect size. For each model, we used the effect size nested in the study ID (1| StudyID/EffectSizeID) as random effects to account for heterogeneity or between-study variations from the primary studies and effect sizes⁶¹. First, we ran models predicting the effects of diversification on crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, and climate regulation individually, without a time moderator (hereafter: moderator), to have reference values for non-temporal syntheses. We then ran the same models by adding the duration of diversification practices as a moderator for both overall and individual agricultural diversification practices. We used the *metafor* r-package⁶² to run the meta-regression models, extracted the model results with the r-package *orchaRd*⁶³, and used *ggplot2* for visualization⁶⁴.

We conducted several tests to detect potential biases in our models. First, we used a funnel plot to visually detect the residuals of the models against the inverse of the standard error (Supplementary Figure 2) with a potential publication bias when the intercept significantly deviated from zero⁶⁵. We ran Egger's regression tests to detect asymmetry of funnel plots (Supplementary Table 5) coupled with fail-safe N analyses based on the Rosenthal method (Supplementary Table 6). We also performed Likelihood

Ratio Tests to detect the statistical significance of publication bias (Supplementary Table 7). We tested the influence of the studies based on the standardized residuals and hat values (Supplementary Figure 3), where the studies with high influence were those with twice the average hat value and with high standardized residuals⁶⁶. Our tests did not reveal any significant issues associated with publication biases or outliers.

To ensure that our results were robust, we performed several sensitivity analyses to compare the outcomes. First, given that we included repeated effect sizes in the response variables, we ran a second-order meta-analysis using a dataset without any repeated responses. Second, since study quality might create some biases, we ran the same analysis but this time with the data set with high-quality studies only. To obtain results from the most conservative analysis, we ran a third type of model with high-quality studies and without repeated responses. These models were run for both the overall effects of diversification and the duration of practices. Overall, the results from these different datasets did not show any significant differences from our main results (i.e., the full dataset, Supplementary Figure 4). For carbon sequestration, excluding repeated variables and/or low-quality studies resulted in neutral long-term effects (>33 years) of diversification, possibly because of data limitations, leading to a wider range of confidence intervals. It is worth noting that all estimates remained positive. Therefore, for coherence of data analyses, we used the full dataset throughout.

We analysed the variation over time in the trade-offs associated with crop yield. We selected studies that tested diversification effects on crop yield and other services at the same time and categorized each effect size as respective, win or lose if the effect size was positive or negative, regardless of their statistical significance. This yielded four standard trade-off outcomes: win_yield - win_Service, win_yield - lose_Service, lose_yield -lose_Service, and lose_yield - win_Service. We then ran hierarchical regression models predicting these outcomes based on a multinomial distribution using the *nnet* r-package⁶⁷ and we used the effect size of the crop yield paired with services, nested within the study ID as random effects. We included the mean of the original comparisons of crop yield and paired services as weights to account for the underlying number of comparisons. We ran the models for crop yield with all the effect size categories combined first and then models

with the individual categories of effect size categories with the duration of the practices as a moderator.

Data and code availability

All data and R code used to conduct the analyses and generate graphs in this study are stored in a private online database accessible only to the authors and are available upon request.

Acknowledgements

We are grateful to Giovanni Tamburini for providing the full texts of the papers used in their studies. E.R. and T.C.W. were funded by the Westlake University start-up fund acquired by T.C.W..

Author contribution

E.R. and T.C.W. developed the study concept. E.R. collected data, conducted all analyses, produced all figures, and wrote the original draft of the manuscript. T.C.W. secured funding and supervised the project. E. R. and T. C. W. edited and revised the manuscript and approved its submission.

Reference:

1. Willett, W. *et al.* Food in the Anthropocene: the EAT–Lancet Commission on healthy diets from sustainable food systems. *The Lancet* **393**, 447–492 (2019).
2. Griggs, D. *et al.* Sustainable development goals for people and planet. *Nature* **495**, 305–307 (2013).
3. Kremen, C., Iles, A. & Bacon, C. Diversified farming systems: An agroecological, systems-based alternative to modern industrial agriculture. *Ecology and Society* **17**, (2012).
4. Hodson, E., Niggli, U., Kitajima, K., Lal, R. & Sadoff, C. *Boost Nature Positive Production - A paper on Action Track 3. United Nations Food Systems Summit 2021* https://sc-fss2021.org/wp-content/uploads/2021/04/Action_Track_3_paper_Boost_Nature_Positive_Production.pdf (2021).
5. Tamburini, G. *et al.* Agricultural diversification promotes multiple ecosystem services without compromising yield. *Sci Adv* **6**, (2020).
6. Rosa-Schleich, J., Loos, J., Musshoff, O. & Tscharntke, T. Ecological-economic trade-offs of Diversified Farming Systems - A review. *ECOLOGICAL ECONOMICS* **160**, 251–263 (2019).
7. Gaupp, F. *et al.* Food system development pathways for healthy, nature-positive and inclusive food systems. *Nat Food* **2**, 928–934 (2021).

8. von Braun, J., Afsana, K., Fresco, L., Hassan, M. & Torero, M. Food systems definition, concept and application for the UN food systems summit. United Nations Food Systems Summit 2021 (2021).
9. FAO. The 10 elements of Agroecology Guiding the transition to sustainable food and agricultural systems. *Fao* 15 (2019).
10. HLPE. Food security and nutrition: building a global narrative towards 2030. A report by the High Level Panel of Experts on Food Security and Nutrition of the Committee on World Food Security, Rome. *High Level Panel of Experts* 112 (2020).
11. Wezel, A. *et al.* Agroecological principles and elements and their implications for transitioning to sustainable food systems. A review. *Agron Sustain Dev* **40**, 40 (2020).
12. Gurr, G. M. *et al.* Multi-country evidence that crop diversification promotes ecological intensification of agriculture. *Nat Plants* **2**, 16014 (2016).
13. He, X. *et al.* Integrating agricultural diversification in China's major policies. *Trends Ecol Evol* **37**, 819–822 (2022).
14. Smith, O. M. *et al.* Organic Farming Provides Reliable Environmental Benefits but Increases Variability in Crop Yields: A Global Meta-Analysis. *Front Sustain Food Syst* **3**, (2019).
15. Muneret, L. *et al.* Evidence that organic farming promotes pest control. *Nat Sustain* **1**, 361–368 (2018).
16. Gattinger, A. *et al.* Enhanced top soil carbon stocks under organic farming. *Proceedings of the National Academy of Sciences* **109**, 18226–18231 (2012).
17. Marrec, R., Brusse, T. & Caro, G. Biodiversity-friendly agricultural landscapes – integrating farming practices and spatiotemporal dynamics. *Trends Ecol Evol* **37**, 731–733 (2022).
18. Tscharrntke, T., Grass, I., Wanger, T. C., Westphal, C. & Batáry, P. Spatiotemporal land-use diversification for biodiversity. *Trends Ecol Evol* **37**, 734–735 (2022).
19. Klapwijk, C. J. *et al.* Analysis of trade-offs in agricultural systems: Current status and way forward. *Current Opinion in Environmental Sustainability* vol. 6 110–115 Preprint at <https://doi.org/10.1016/j.cosust.2013.11.012> (2014).
20. He, X. *et al.* Agricultural diversification promotes sustainable and resilient global rice production. *Nat Food* **4**, 788–796 (2023).
21. Toledo-Hernández, M. *et al.* Genome-edited tree crops: mind the socioeconomic implementation gap. *Trends Ecol Evol* **36**, 972–975 (2021).
22. Niggli, U., Sonneveld, M. & Kummer, S. Pathways to Advance Agroecology for a Successful Transformation to Sustainable Food Systems. *Science and Innovations for Food Systems Transformation* 341–359 (2023) doi:10.1007/978-3-031-15703-5_18.
23. Mehrabi, Z. *et al.* Research priorities for global food security under extreme events. *One Earth* **5**, 756–766 (2022).
24. Ruckelshaus, M. H. *et al.* The IPBES Global Assessment: Pathways to Action. *Trends Ecol Evol* **35**, 407–414 (2020).
25. IPBES. Summary for policymakers of the global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. (2019).
26. Qi, G. *et al.* Yield Responses of Wheat to Crop Residue Returning in China: A Meta-Analysis. *Crop Sci* **59**, 2185–2200 (2019).
27. Renard, D. & Tilman, D. National food production stabilized by crop diversity. *Nature* **571**, 257–260 (2019).
28. WWF. Bending the curve of biodiversity loss. Living Planet Report 2020 (2020).
29. Dainese, M. *et al.* A global synthesis reveals biodiversity-mediated benefits for crop production. *Sci Adv* **5**, 1–14 (2019).
30. Klein, A.-M. *et al.* Importance of pollinators in changing landscapes for world crops. *Proceedings of the Royal Society B: Biological Sciences* **274**, 303–313 (2007).

31. Tscharntke, T. *et al.* When natural habitat fails to enhance biological pest control – Five hypotheses. *Biol Conserv* **204**, 449–458 (2016).
32. Bai, X. *et al.* Responses of soil carbon sequestration to climate-smart agriculture practices: A meta-analysis. *Glob Chang Biol* **25**, 2591–2606 (2019).
33. Lessmann, M., Ros, G. H., Young, M. D. & Vries, W. Global variation in soil carbon sequestration potential through improved cropland management. *Glob Chang Biol* **28**, 1162–1177 (2022).
34. Searchinger, T. *et al.* Creating a sustainable food future: A menu of solutions to feed more than 9 billion people by 2050. *World Resources Institute* 1–5 (2014).
35. Leclère, D. *et al.* Bending the curve of terrestrial biodiversity needs an integrated strategy. *Nature* **585**, 551–556 (2020).
36. Qin, Z., Dunn, J. B., Kwon, H., Mueller, S. & Wander, M. M. Soil carbon sequestration and land use change associated with biofuel production: empirical evidence. *GCB Bioenergy* **8**, 66–80 (2016).
37. Guenet, B. *et al.* Can N₂O emissions offset the benefits from soil organic carbon storage? *Glob Chang Biol* **27**, 237–256 (2021).
38. Chen, Y., Camps-Arbestain, M., Shen, Q., Singh, B. & Cayuela, M. L. The long-term role of organic amendments in building soil nutrient fertility: a meta-analysis and review. *Nutr Cycl Agroecosyst* **111**, 103–125 (2018).
39. Ledo, A. *et al.* Changes in soil organic carbon under perennial crops. *Glob Chang Biol* **26**, 4158–4168 (2020).
40. Wei, Z., Hoffland, E., Zhuang, M., Hellegers, P. & Cui, Z. Organic inputs to reduce nitrogen export via leaching and runoff: A global meta-analysis. *Environmental Pollution* **291**, 118176 (2021).
41. Tesfaye, F. *et al.* Could biochar amendment be a tool to improve soil availability and plant uptake of phosphorus? A meta-analysis of published experiments. *Environmental Science and Pollution Research* **28**, 34108–34120 (2021).
42. Sánchez, A. C., Kamau, H. N., Grazioli, F. & Jones, S. K. Financial profitability of diversified farming systems: A global meta-analysis. *Ecological Economics* **201**, 107595 (2022).
43. Smith, M. E. *et al.* Increasing crop rotational diversity can enhance cereal yields. *Commun Earth Environ* **4**, 89 (2023).
44. Pumariño, L. *et al.* Effects of agroforestry on pest, disease and weed control: A meta-analysis. *Basic Appl Ecol* **16**, 573–582 (2015).
45. Niether, W., Jacobi, J., Blaser, W. J., Andres, C. & Armengot, L. Cocoa agroforestry systems versus monocultures: a multi-dimensional meta-analysis. *Environmental Research Letters* **15**, 104085 (2020).
46. Tscharntke, T., Grass, I., Wanger, T. C., Westphal, C. & Batáry, P. Beyond organic farming – harnessing biodiversity-friendly landscapes. *Trends Ecol Evol* **36**, 919–930 (2021).
47. Cui, Z. *et al.* Pursuing sustainable productivity with millions of smallholder farmers. *Nature* **555**, 363–366 (2018).
48. Hertel, T. W., Elouafi, I., Ewert, F. & Tanticharoen, morakot. Building Resilience to Vulnerabilities, Shocks and Stresses. *UN Food Systems Summit* 1–24 (2021).
49. FAO, European Union & CIRAD. Catalysing the sustainable and inclusive transformation of food systems. (2021).
50. Egli, L., Meyer, C., Scherber, C., Kreft, H. & Tscharntke, T. Winners and losers of national and global efforts to reconcile agricultural intensification and biodiversity conservation. *Glob Chang Biol* **24**, 2212–2228 (2018).
51. Martinez-Salinas, A. *et al.* Interacting pest control and pollination services in coffee systems. *Proc Natl Acad Sci U S A* **119**, 1–7 (2022).

52. Johnston, A. E. & Poulton, P. R. The importance of long-term experiments in agriculture: their management to ensure continued crop production and soil fertility; the Rothamsted experience. *Eur J Soil Sci* **69**, 113–125 (2018).
53. Ortiz-Bobea, A., Ault, T. R., Carrillo, C. M., Chambers, R. G. & Lobell, D. B. Anthropogenic climate change has slowed global agricultural productivity growth. *Nat Clim Chang* **11**, 306–312 (2021).
54. Aggarwal, P., Vyas, S., Thornton, P. & Campbell, B. M. How much does climate change add to the challenge of feeding the planet this century? *Environmental Research Letters* **14**, (2019).
55. Montenegro, M. & Wit, D. Operating principles for collective scholar-activism : Early insights from the Agroecology Research-Action Collective. **10**, 319–337 (2021).
56. Holt, A. R., Alix, A., Thompson, A. & Maltby, L. Science of the Total Environment Food production , ecosystem services and biodiversity : We can ' t have it all everywhere. *Science of the Total Environment, The* **573**, 1422–1429 (2016).
57. Hegwood, M., Langendorf, R. E. & Burgess, M. G. Why win–wins are rare in complex environmental management. *Nat Sustain* **5**, 674–680 (2022).
58. Kuyah, S. *et al.* Agroforestry delivers a win-win solution for ecosystem services in sub-Saharan Africa. A meta-analysis. *Agron Sustain Dev* **39**, (2019).
59. Mo, L. *et al.* Integrated global assessment of the natural forest carbon potential. *Nature* **624**, (2023).
60. IPBES. Methodological Assessment Report on the Diverse Values and Valuation of Nature of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. (IPBES secretariat, Bonn, Germany., 2022). doi:<https://doi.org/10.5281/zenodo.6522522>.
61. Schmidt, F. L. & Oh, I. S. Methods for second order meta-analysis and illustrative applications. *Organ Behav Hum Decis Process* **121**, 204–218 (2013).
62. Viechtbauer, W. Conducting meta-analyses in R with the metafor. *J Stat Softw* **36**, 1–48 (2010).
63. Nakagawa, S. *et al.* orchaRd 2.0: An R package for visualising meta-analyses with orchard plots. *Methods Ecol Evol* **2023**, 1–8 (2023).
64. Wickham, H. *ggplot2: Elegant Graphics for Data Analysis*. (Springer-Verlag New York, 2016).
65. Viechtbauer, W. Model Checking in Meta-Analysis. in *Handbook of Meta-Analysis* (eds. Schmid, C. H., Stijnen, T. & White, I.) 219–254 (Chapman and Hall/CRC, 2020).
66. Marks-Anglin, A., Duan, R., Chen, Y., Panagiotou, O. & Schmid, C. H. Publication and Outcome Reporting Bias. in *Handbook of Meta-Analysis* (eds. Schmid, C. H., Stijnen, T. & White, I. R.) (Chapman and Hall/CRC, 2020).
67. Brooks, M. E. *et al.* glmmTMB Balances Speed and Flexibility Among Packages for Zero-inflated Generalized Linear Mixed Modeling. *R J* **9**, 378–400 (2017).

Title: Decades matter: Agricultural diversification increases financial profitability, biodiversity, and ecosystem services over time.

Authors: Estelle Raveloaritiana^{1,2,*}, Thomas Cherico Wanger^{1,2,3,4,5*}

¹ Sustainable Agricultural Systems & Engineering Laboratory, School of Engineering, Westlake University, Hangzhou, China.

² Key Laboratory of Coastal Environment and Resources of Zhejiang Province, Westlake University, Hangzhou, China.

³ Faculty of Science, Technology & Medicine, University of Luxembourg, Luxembourg

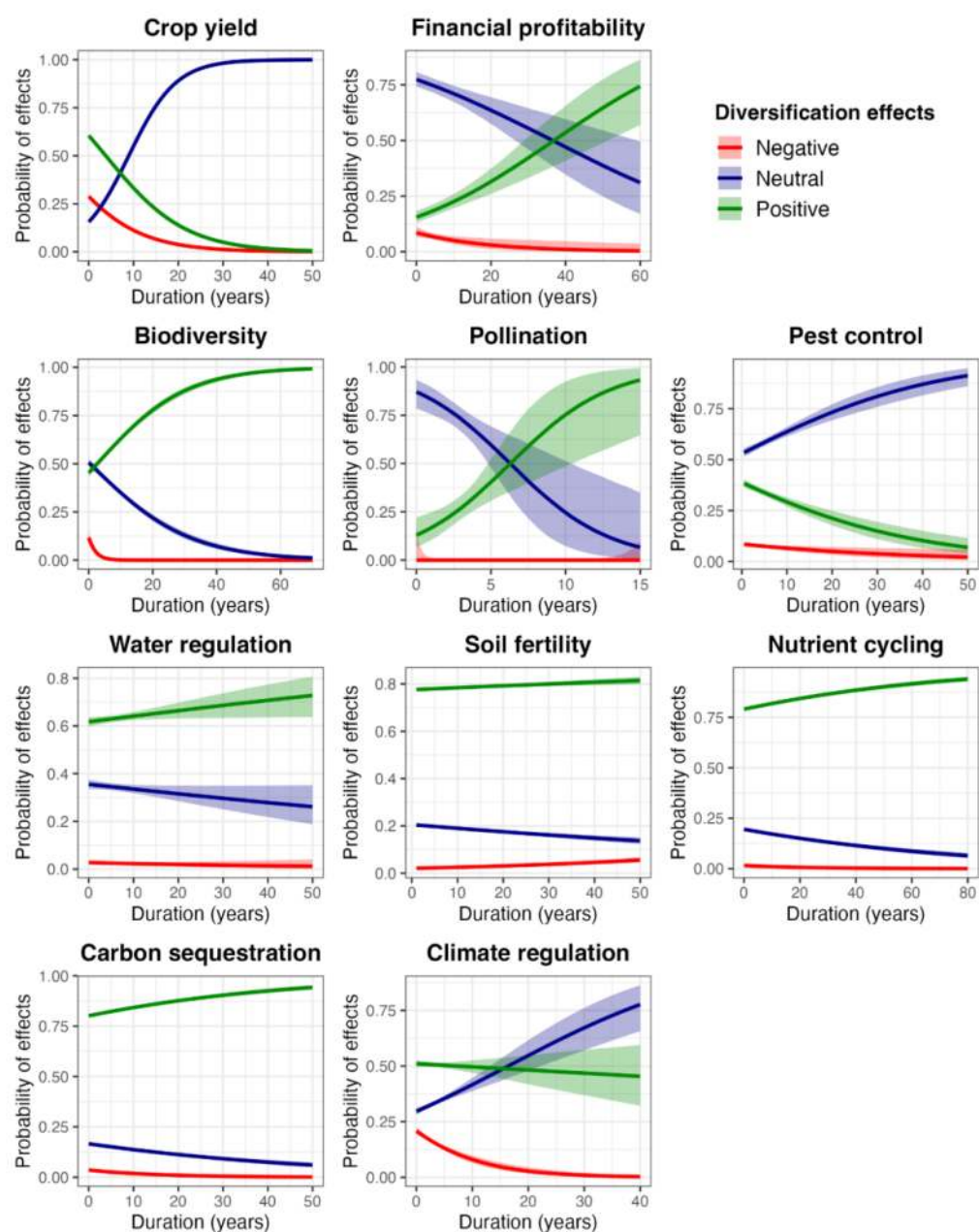
⁴ GlobalAgroforestryNetwork.org, Hangzhou, China

⁵ ChinaRiceNetwork.org, Hangzhou, China

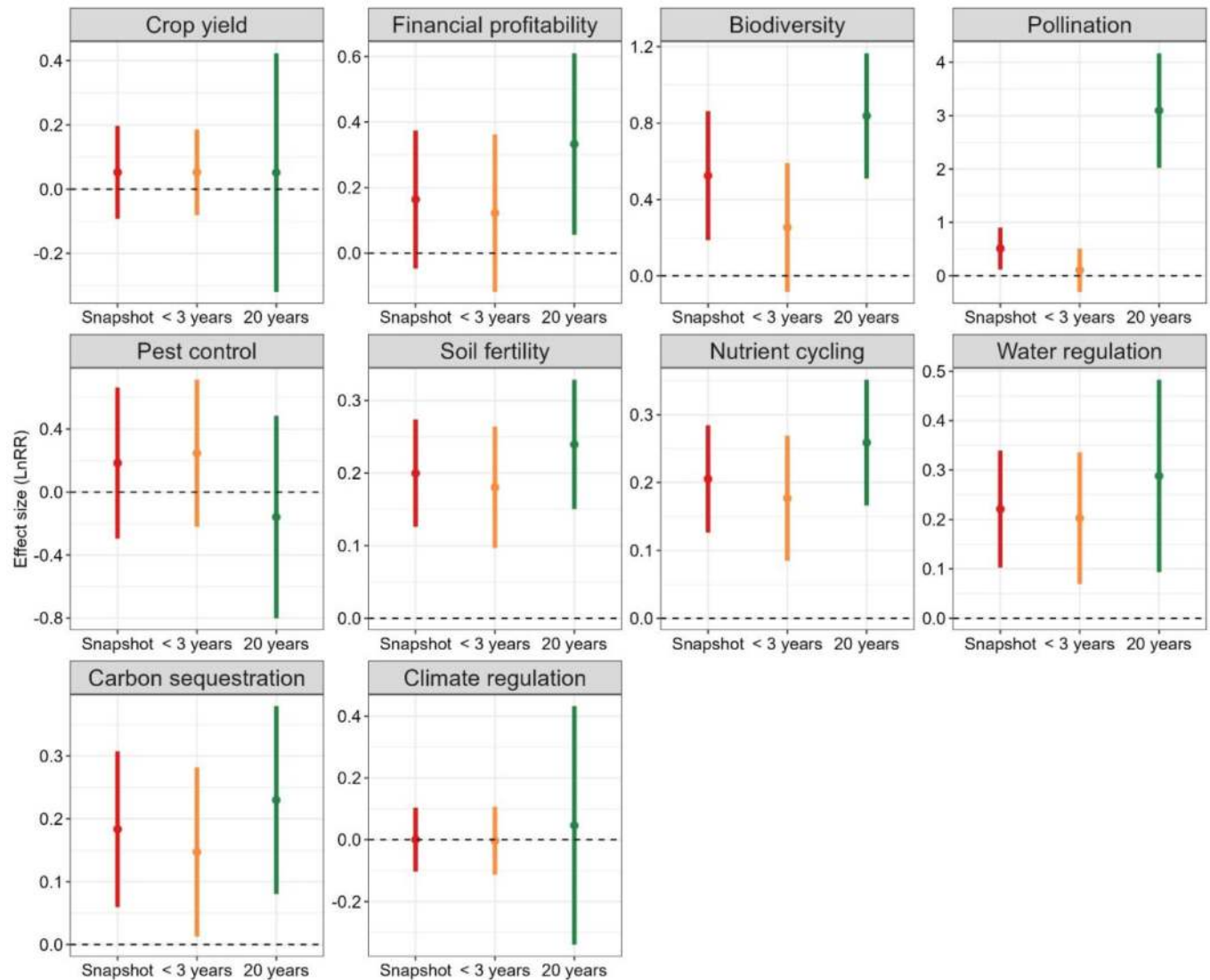
*Corresponding authors: eraveloaritiana@gmail.com (ER); tomcwanger@gmail.com (TCW)

Extended Data and Supplementary materials

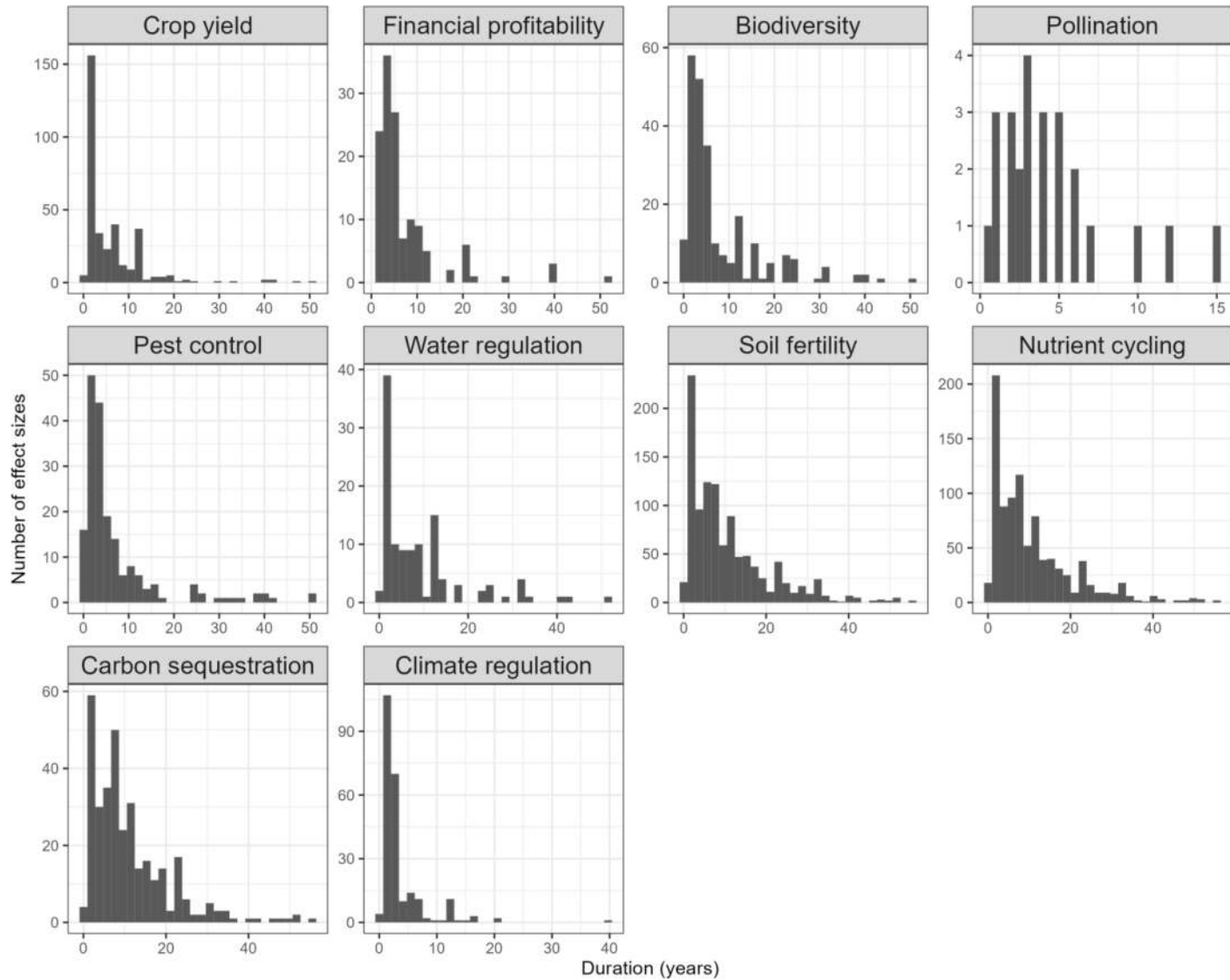
****Extended data**



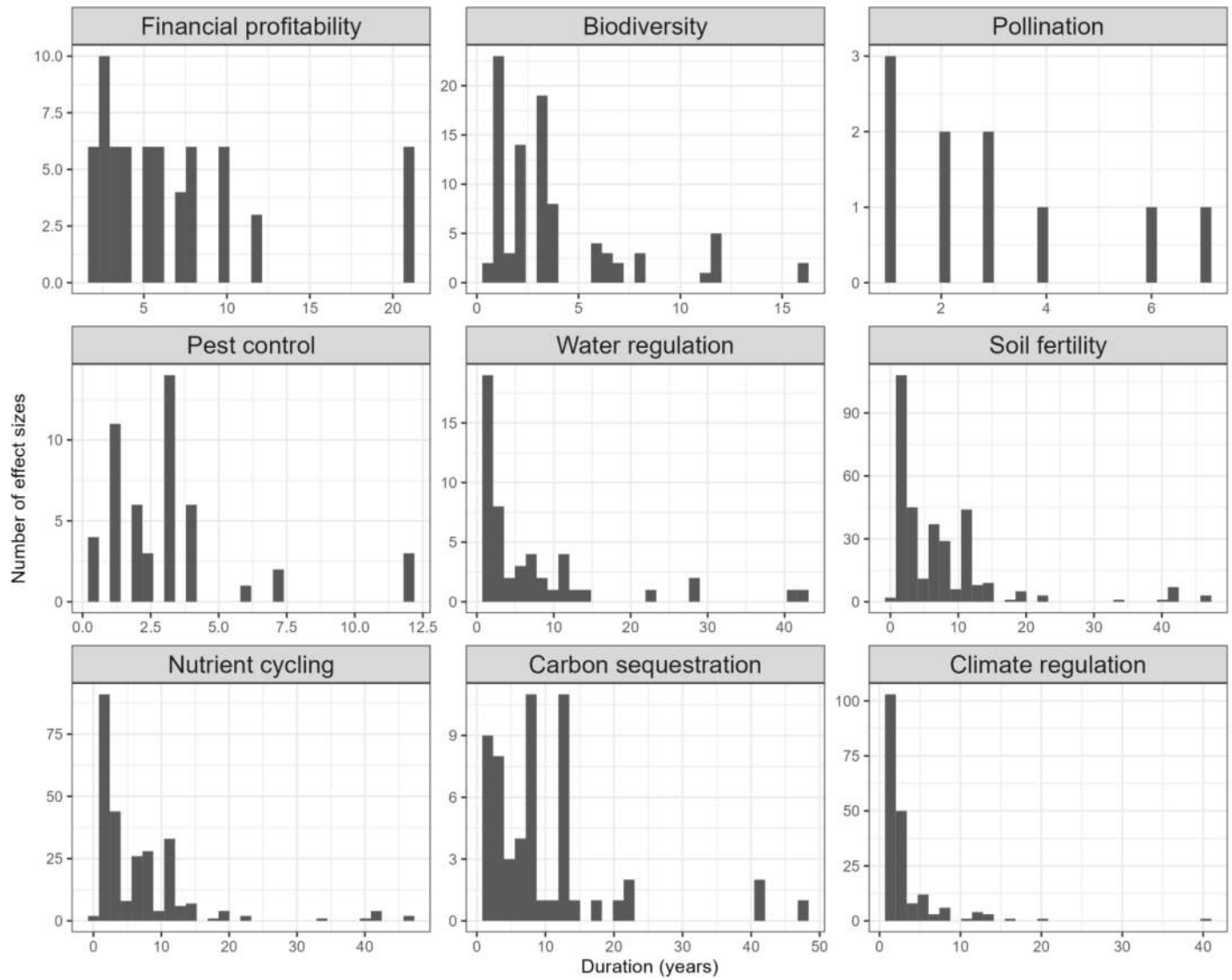
Extended Data Fig. 1 | Quantitative systematic review of the effects of the diversification practices on finance, biodiversity, and ecosystem services



Extended Data Fig. 2| Snapshot, short-term (<3 years) and long-term (20 years) effects of diversification practices on socioeconomic, biological community, soil quality, and climate change mitigation related variables



Extended Data Fig. 3| Vote count of the effect sizes included in the effects of agricultural diversification over time



Extended Data Fig. 4| Vote count of the effect sizes included in the trade-off analysis between crop yield and other services across the duration of the diversification practices.

Extended Data Table 1. Estimates, standard errors, p-values and r-squared of the response variables depict the effects of overall diversification practices without and with duration as moderator.

No moderators						
	Moderators	Estimate	SE	z-value	p-value	
Crop yield	-	0.0524	0.0737	0.7108	0.4772	
Financial profitability	-	0.1639	0.1074	1.5258	0.1271	
Biodiversity	-	0.5247	0.1722	3.0465	0.0023	
Pollination	-	0.5125	0.2005	2.5553	0.0106	
Pest control	-	0.1845	0.2449	0.7532	0.4513	
Soil fertility	-	0.1998	0.0377	5.2938	<.0001	
Nutrient cycling	-	0.2053	0.0403	5.0944	<.0001	
Water regulation	-	0.221	0.0604	3.6617	0.0003	
Carbon sequestration	-	0.1836	0.0631	2.9096	0.0036	
Climate regulation	-	0.0004	0.0528	0.008	0.9936	
With Duration as moderator						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R ²
Crop yield	Intercept	0.0526	0.0668	0.7863	0.4317	0.000
	Duration (years)	0	0.0089	-0.0053	0.9957	
Financial profitability	Intercept	0.1024	0.1245	0.8228	0.4106	0.128
	Duration (years)	0.0113	0.0052	2.1796	0.0293	
Biodiversity	Intercept	0.1984	0.1794	1.1062	0.2687	0.204
	Duration (years)	0.031	0.0083	3.7512	0.0002	
Pollination	Intercept	-0.6172	0.2781	-2.2195	0.0265	0.981
	Duration (years)	0.3548	0.068	5.2171	<.0001	
Pest control	Intercept	0.2947	0.2339	1.2597	0.2078	0.146
	Duration (years)	-0.022	0.0093	-2.3623	0.0182	
Soil fertility	Intercept	0.1739	0.0452	3.8486	0.0001	0.017
	Duration (years)	0.0032	0.0025	1.2884	0.1976	
Nutrient cycling	Intercept	0.1679	0.0498	3.3716	0.0007	0.030
	Duration (years)	0.0044	0.0026	1.7261	0.0843	
Water regulation	Intercept	0.1944	0.0735	2.6456	0.0082	0.035
	Duration (years)	0.0046	0.0059	0.7689	0.4419	
Carbon sequestration	Intercept	0.1384	0.0716	1.934	0.031	0.025
	Duration (years)	0.0045	0.0036	1.2395	0.2151	
Climate regulation	Intercept	-0.0079	0.0641	-0.1227	0.9023	0.002
	Duration (years)	0.0026	0.0109	0.241	0.8095	

Extended Data Table 2. Estimates of non-temporal effects sizes as well as the short term (<3 years) and long-term (20 years) estimates and the percentage differences amongst the estimates.

	Non-temporal Estimate	Estimate 0 to 3 years	estimate 20 years	Percentage difference between non-temporal and < 3 years	Percentage difference between non-temporal and 20 years estimates	Percentage difference between < 3 years and 20 years estimates
Crop yield	0.0524	0.0525	0.0526	0	0	0
Financial profitability	0.1639	0.1222	0.3327	-25	100	172
Biodiversity	0.5247	0.2534	0.8377	-52	56	231
Pollination**	0.5125	0.1059	3.0939	-79	1164	2823
Pest control	0.1845	0.2463	-0.1588	33	-179	-164
Soil fertility	0.1998	0.1803	0.2394	-10	19	33
Nutrient cycling	0.2053	0.1768	0.2589	-14	25	46
Water regulation	0.2210	0.2026	0.2881	-8	30	42
Carbon sequestration	0.1836	0.1474	0.2300	-20	24	56
Climate regulation	0.0004	0.0001	0.0462	-75	10925	46103

**10 years for Pollination

Extended Data Table 3. Estimates, standard errors, and p-values of the response variables showing the effects of different categories of diversification practices on all response variables without temporal consideration.

Crop diversification				
	Estimate	SE	z-value	p-value
Crop yield	0.0365	0.0832	0.4391	0.6606
Financial profitability	0.1548	0.114	1.3579	0.1745
Biodiversity	0.2427	0.2954	0.8216	0.4113
Pollination	0.1816	0.1532	1.1852	0.2359
Pest control	0.2848	0.2792	1.0199	0.3078
Water regulation	0.5619	0.4344	1.2935	0.1958
Soil fertility	0.1731	0.0706	2.4521	0.0142
Nutrient cycling	0.1932	0.075	2.5768	0.01
Carbon sequestration	0.1558	0.1323	1.1777	0.2389
Climate regulation	0.2238	0.2112	1.0597	0.2893
Non-crop diversification				
	Estimate	SE	z-value	p-value
Crop yield	0.2783	0.149	1.8679	0.0618
Financial profitability	-0.0335	0.4925	-0.068	0.9458
Biodiversity	0.5212	0.0722	7.2203	<.0001

Pollination	1.1824	0.373	3.1705	0.0015
Pest control	0.1851	0.0518	3.5709	0.0004
Water regulation	0.477	0.2618	1.8223	0.0684
Soil fertility	0.2233	0.078	2.8614	0.0042
Nutrient cycling	0.2569	0.1176	2.184	0.029
Carbon sequestration	0.3652	0.0795	4.5923	<.0001
Climate regulation	-0.0756	0.0863	-0.8768	0.3806
Reduced tillage				
	Estimate	SE	z-value	p-value
Crop yield	-0.0276	0.0232	-1.1893	0.2343
Biodiversity	0.4037	0.1904	2.1199	0.034
Water regulation	0.0896	0.1042	0.8601	0.3898
Soil fertility	0.1539	0.062	2.4812	0.0131
Nutrient cycling	0.1402	0.0578	2.4274	0.0152
Carbon sequestration	0.1014	0.0108	9.3981	<.0001
Climate regulation	0.0236	0.0997	0.2366	0.8129
Organic amendment				
	Estimate	SE	z-value	p-value
Crop yield	0.1105	0.0287	3.8512	0.0001
Biodiversity	0.7264	0.1916	3.7905	0.0002
Water regulation	0.2151	0.0632	3.4023	0.0007
Soil fertility	0.2139	0.0702	3.0488	0.0023
Nutrient cycling	0.223	0.0638	3.4951	0.0005
Carbon sequestration	0.2013	0.0874	2.3022	0.0213
Climate regulation	-0.0114	0.0773	-0.1476	0.8827
Organic farming				
	Estimate	SE	z-value	p-value
Crop yield	-0.2027	0.1719	-1.1788	0.2385
Financial profitability	0.1821	0.1477	1.2329	0.2176
Biodiversity	0.612	0.578	1.0589	0.2897
Pollination	0.1959	0.0658	2.9751	0.0029
Pest control	-0.0685	0.1946	-0.3519	0.7249
Water regulation	0.27	0.22	1.2274	0.2197
Soil fertility	0.3866	0.1421	2.7199	0.0065
Nutrient cycling	0.3847	0.177	2.1738	0.0297
Carbon sequestration	0.8812	0.5559	1.5853	0.1129
Inoculation				
	Estimate	SE	z-value	p-value
Crop yield	0.1533	0.1782	0.8602	0.3897
Soil fertility	0.3382	0.1605	2.107	0.0351
Nutrient cycling	0.1593	0.0601	2.651	0.008
Climate regulation	0.0536	0.1153	0.4647	0.6421

Extended Data Table 4. Estimates, standard errors, p-values and r-squared of the response variables showing the effects of different categories of diversification practices on all response variables over time.

Crop diversification						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R ²
Crop yield	Intercept	0.037	0.103	0.361	0.718	0.000
	Duration (years)	0.000	0.009	-0.023	0.982	
Financial profitability	Intercept	0.090	0.154	0.583	0.560	0.281
	Duration (years)	0.011	0.005	2.477	0.013	
Biodiversity	Intercept	0.165	0.298	0.554	0.580	0.083
	Duration (years)	0.009	0.008	1.089	0.276	
Pollination	Intercept	0.527	0.736	0.716	0.474	0.111
	Duration (years)	-0.104	0.119	-0.874	0.382	
Pest control	Intercept	0.408	0.270	1.510	0.131	0.224
	Duration (years)	-0.024	0.011	-2.313	0.021	
Soil fertility	Intercept	0.201	0.091	2.217	0.027	0.007
	Duration (years)	-0.002	0.004	-0.463	0.644	
Nutrient cycling	Intercept	0.228	0.098	2.322	0.020	0.008
	Duration (years)	-0.002	0.004	-0.550	0.582	
Water regulation	Intercept	0.897	0.605	1.482	0.138	0.276
	Duration (years)	-0.074	0.076	-0.974	0.330	
Carbon sequestration	Intercept	0.187	0.170	1.103	0.270	0.005
	Duration (years)	-0.002	0.008	-0.282	0.778	
Climate regulation	Intercept	0.113	0.456	0.248	0.804	0.010
	Duration (years)	0.086	0.285	0.301	0.764	
Non-crop diversification						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R ²
Crop yield	Intercept	-0.395	0.254	-1.555	0.120	0.668
	Duration (years)	0.132	0.051	2.616	0.009	
Biodiversity	Intercept	0.359	0.153	2.344	0.019	0.547
	Duration (years)	0.043	0.013	3.411	0.001	
Pollination	Intercept	-0.271	0.513	-0.527	0.598	0.998
	Duration (years)	0.401	0.083	4.851	<.0001	
Pest control	Intercept	0.038	0.093	0.405	0.685	0.314
	Duration (years)	0.050	0.030	1.647	0.100	
Soil fertility	Intercept	0.070	0.094	0.748	0.454	0.486
	Duration (years)	0.014	0.005	2.740	0.006	
Nutrient cycling	Intercept	0.081	0.180	0.451	0.652	0.287
	Duration (years)	0.014	0.009	1.650	0.099	
Water regulation	Intercept	0.4144	0.2651	1.5634	0.118	0.018
	Duration (years)	0.0036	0.0103	0.3543	0.7231	
Carbon sequestration	Intercept	0.0928	0.043	2.16	0.0308	0.896
	Duration (years)	0.0156	0.0024	6.5079	<.0001	
Reduced tillage						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R ²
Crop yield	Intercept	-0.039	0.024	-1.639	0.101	0.032

	Duration (years)	0.002	0.002	1.112	0.266	
Biodiversity	Intercept	0.327	0.256	1.280	0.201	0.032
	Duration (years)	0.008	0.011	0.671	0.503	
Soil fertility	Intercept	0.165	0.059	2.811	0.005	0.002
	Duration (years)	-0.001	0.003	-0.319	0.750	
Nutrient cycling	Intercept	0.126	0.047	2.690	0.007	0.004
	Duration (years)	0.001	0.003	0.410	0.682	
Water regulation	Intercept	0.103	0.105	0.980	0.327	0.006
	Duration (years)	-0.002	0.009	-0.237	0.813	
Carbon sequestration	Intercept	0.077	0.015	5.115	<.0001	0.580
	Duration (years)	0.002	0.001	2.154	0.031	
Climate regulation	Intercept	0.056	0.098	0.574	0.566	0.006
	Duration (years)	-0.006	0.011	-0.518	0.605	
Organic amendment						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R ²
Crop yield	Intercept	0.077	0.035	2.235	0.025	0.344
	Duration (years)	0.010	0.006	1.702	0.089	
Biodiversity	Intercept	0.141	0.095	1.482	0.138	0.891
	Duration (years)	0.041	0.004	10.719	<.0001	
Soil fertility	Intercept	0.185	0.079	2.338	0.019	0.036
	Duration (years)	0.005	0.005	1.168	0.243	
Nutrient cycling	Intercept	0.165	0.072	2.279	0.023	0.142
	Duration (years)	0.010	0.004	2.668	0.008	
Water regulation	Intercept	0.221	0.069	3.211	0.001	0.022
	Duration (years)	-0.002	0.008	-0.247	0.805	
Carbon sequestration	Intercept	0.166	0.104	1.593	0.111	0.035
	Duration (years)	0.006	0.007	0.916	0.360	
Climate regulation	Intercept	-0.030	0.085	-0.353	0.724	0.012
	Duration (years)	0.007	0.018	0.392	0.695	
Organic farming						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R ²
Crop yield	Intercept	-0.223	0.225	-0.987	0.324	0.008
	Duration (years)	0.003	0.012	0.234	0.815	
Financial profitability	Intercept	0.116	0.196	0.592	0.554	0.044
	Duration (years)	0.014	0.024	0.577	0.564	
Biodiversity	Intercept	0.037	0.689	0.053	0.958	0.064
	Duration (years)	0.039	0.026	1.509	0.131	
Pollination	Intercept	0.208	0.259	0.805	0.421	0.009
	Duration (years)	-0.006	0.099	-0.055	0.956	
Pest control	Intercept	0.023	0.184	0.124	0.901	0.237
	Duration (years)	-0.016	0.016	-1.043	0.297	
Soil fertility	Intercept	0.4611	0.2507	1.8397	0.0658	0.029
	Duration (years)	-0.0064	0.0156	-0.414	0.6789	
Nutrient cycling	Intercept	0.394	0.250	1.575	0.115	0.0003
	Duration (years)	-0.001	0.020	-0.052	0.959	
Carbon sequestration	Intercept	1.775	0.317	5.600	<.0001	0.87
	Duration (years)	-0.050	0.026	-1.906	0.057	

Inoculation						
Response variables	Moderators	Estimate	SE	z-value	p-value	Marginal R²
Crop yield	Intercept	0.067	0.228	0.293	0.769	0.031
	Duration (years)	0.051	0.060	0.855	0.392	
Soil fertility	Intercept	0.617	0.225	2.747	0.006	0.485
	Duration (years)	-0.187	0.119	-1.574	0.115	
Nutrient cycling	Intercept	0.050	0.071	0.699	0.485	0.454
	Duration (years)	0.063	0.022	2.816	0.005	
Climate regulation	Intercept	0.032	0.120	0.267	0.790	0.008
	Duration (years)	0.015	0.022	0.694	0.488	

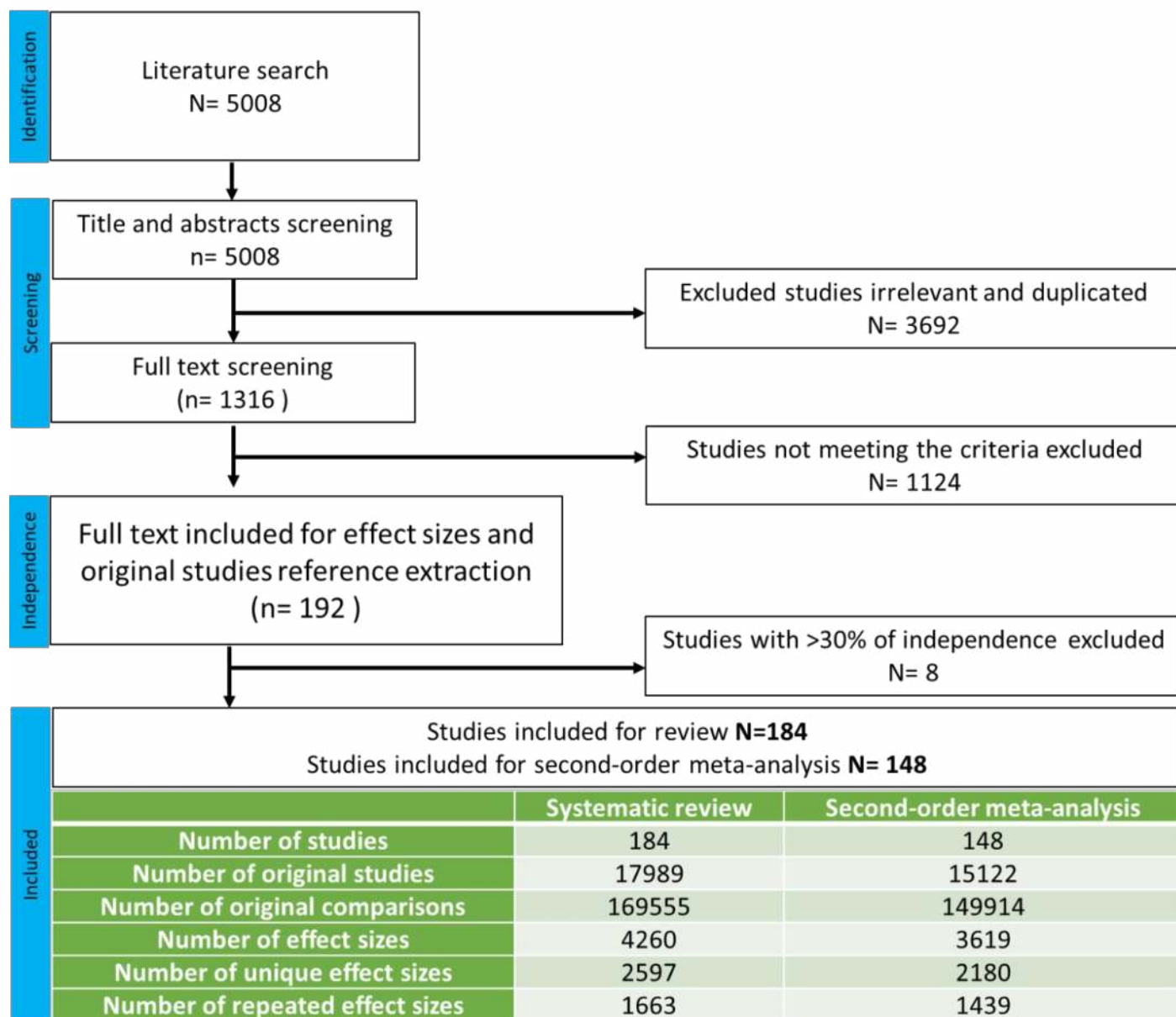
Extended Data Table 5. Result of the multinomial regression models predicting the probability of trade-off outcomes of agricultural diversification practice over time, with crop yield vs. other response variables, in overall and individually.

Categories of response variable	Response level	Predictors	Log-Odds	std.error	statistic	p.value	RRR	nobs
Yield vs. all services	neg-pos	(Intercept)	0.566	0.015	38.813	0.000	1.762	1448
Yield vs. all services	neg-pos	Duration	0.098	0.005	17.839	0.000	1.103	1448
Yield vs. all services	pos-neg	(Intercept)	0.647	0.016	39.246	0.000	1.909	1448
Yield vs. all services	pos-neg	Duration	-0.174	0.008	-20.521	0.000	0.841	1448
Yield vs. all services	pos-pos	(Intercept)	1.685	0.013	125.610	0.000	5.390	1448
Yield vs. all services	pos-pos	Duration	0.006	0.005	1.112	0.266	1.006	1448
Yield vs. Financial profitability	neg-pos	(Intercept)	0.827	0.049	16.976	0.000	2.287	98
Yield vs. Financial profitability	neg-pos	Duration	-0.097	0.015	-6.257	0.000	0.908	98
Yield vs. Financial profitability	pos-neg	(Intercept)	0.380	0.075	5.073	0.000	1.463	98
Yield vs. Financial profitability	pos-neg	Duration	-0.228	0.038	-6.051	0.000	0.796	98
Yield vs. Financial profitability	pos-pos	(Intercept)	1.688	0.061	27.767	0.000	5.407	98
Yield vs. Financial profitability	pos-pos	Duration	-0.446	0.033	-13.711	0.000	0.640	98
Yield vs. Biodiversity	neg-pos	(Intercept)	-0.461	0.083	-5.543	0.000	0.631	93
Yield vs. Biodiversity	neg-pos	Duration	0.890	0.095	9.357	0.000	2.436	93
Yield vs. Biodiversity	pos-neg	(Intercept)	-1.687	0.148	-11.411	0.000	0.185	93
Yield vs. Biodiversity	pos-neg	Duration	0.721	0.113	6.404	0.000	2.055	93
Yield vs. Biodiversity	pos-pos	(Intercept)	-0.107	0.081	-1.329	0.184	0.898	93
Yield vs. Biodiversity	pos-pos	Duration	0.902	0.095	9.501	0.000	2.465	93
Yield vs. Pollination	neg-pos	(Intercept)	-1.887	0.371	-5.093	0.000	0.151	11
Yield vs. Pollination	neg-pos	Duration	1.342	0.188	7.128	0.000	3.828	11
Yield vs. Pollination	pos-neg	(Intercept)	-3.758	0.541	-6.952	0.000	0.023	11
Yield vs. Pollination	pos-neg	Duration	1.597	0.226	7.066	0.000	4.936	11
Yield vs. Pollination	pos-pos	(Intercept)	-101.12	135.575	-0.746	0.456	0.000	11
Yield vs. Pollination	pos-pos	Duration	1.787	0.920	0.802	0.422	5.973	11
Yield vs. Pest control	neg-pos	(Intercept)	0.715	0.054	13.323	0.000	2.045	55
Yield vs. Pest control	neg-pos	Duration	-0.106	0.026	-4.028	0.000	0.899	55
Yield vs. Pest control	pos-neg	(Intercept)	-1.127	0.119	-9.492	0.000	0.324	55
Yield vs. Pest control	pos-neg	Duration	0.136	0.039	3.515	0.000	1.146	55
Yield vs. Pest control	pos-pos	(Intercept)	0.182	0.066	2.778	0.005	1.200	55
Yield vs. Pest control	pos-pos	Duration	-0.098	0.034	-2.908	0.004	0.906	55
Yield vs. Water regulation	neg-pos	(Intercept)	0.362	0.051	7.134	0.000	1.436	72
Yield vs. Water regulation	neg-pos	Duration	-0.033	0.023	-1.471	0.141	0.967	72

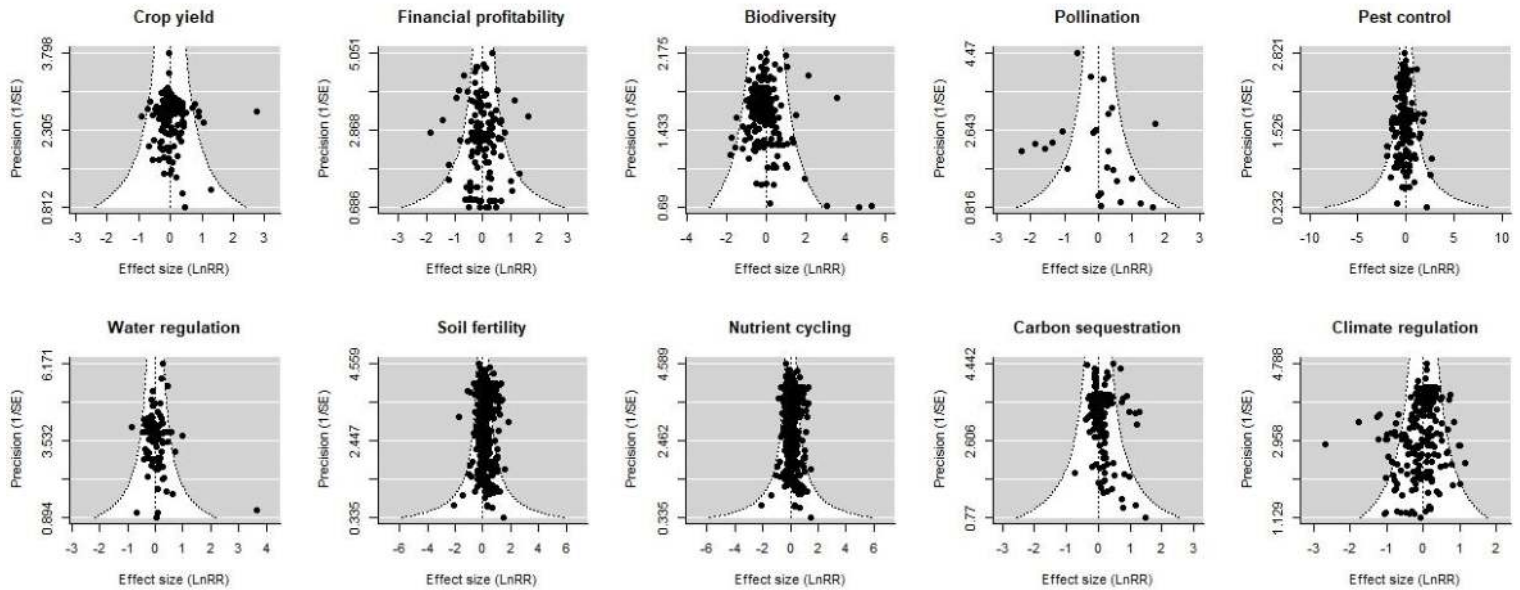
Yield vs. Water regulation	pos-neg	(Intercept)	0.300	0.052	5.739	0.000	1.349	72
Yield vs. Water regulation	pos-neg	Duration	-0.037	0.024	-1.569	0.117	0.964	72
Yield vs. Water regulation	pos-pos	(Intercept)	1.524	0.041	37.022	0.000	4.589	72
Yield vs. Water regulation	pos-pos	Duration	-0.010	0.017	-0.602	0.547	0.990	72
Yield vs. Soil fertility	neg-pos	(Intercept)	1.372	0.051	26.647	0.000	3.942	421
Yield vs. Soil fertility	neg-pos	Duration	-0.004	0.011	-0.388	0.698	0.996	421
Yield vs. Soil fertility	pos-neg	(Intercept)	1.889	0.053	35.354	0.000	6.611	421
Yield vs. Soil fertility	pos-neg	Duration	-0.431	0.019	-23.110	0.000	0.650	421
Yield vs. Soil fertility	pos-pos	(Intercept)	2.846	0.050	56.623	0.000	17.219	421
Yield vs. Soil fertility	pos-pos	Duration	-0.133	0.011	-12.535	0.000	0.876	421
Yield vs. Nutrient cycling	neg-pos	(Intercept)	1.557	0.057	27.435	0.000	4.746	357
Yield vs. Nutrient cycling	neg-pos	Duration	-0.010	0.012	-0.844	0.399	0.990	357
Yield vs. Nutrient cycling	pos-neg	(Intercept)	1.334	0.064	20.796	0.000	3.796	357
Yield vs. Nutrient cycling	pos-neg	Duration	-0.373	0.028	-13.495	0.000	0.689	357
Yield vs. Nutrient cycling	pos-pos	(Intercept)	2.878	0.056	51.574	0.000	17.773	357
Yield vs. Nutrient cycling	pos-pos	Duration	-0.122	0.012	-9.901	0.000	0.885	357
Yield vs. Carbon sequestration	pos-neg	(Intercept)	0.005	0.055	0.083	0.934	1.005	72
Yield vs. Carbon sequestration	pos-neg	Duration	-0.511	0.045	-11.256	0.000	0.600	72
Yield vs. Carbon sequestration	pos-pos	(Intercept)	0.834	0.019	42.850	0.000	2.302	72
Yield vs. Carbon sequestration	pos-pos	Duration	-0.100	0.004	-22.594	0.000	0.905	72
Yield vs. Climate regulation	neg-pos	(Intercept)	0.164	0.031	5.334	0.000	1.178	269
Yield vs. Climate regulation	neg-pos	Duration	-0.115	0.020	-5.806	0.000	0.891	269
Yield vs. Climate regulation	pos-neg	(Intercept)	0.322	0.026	12.419	0.000	1.379	269
Yield vs. Climate regulation	pos-neg	Duration	-0.042	0.015	-2.863	0.004	0.959	269
Yield vs. Climate regulation	pos-pos	(Intercept)	0.839	0.022	38.336	0.000	2.314	269
Yield vs. Climate regulation	pos-pos	Duration	-0.012	0.012	-1.033	0.302	0.988	269

****Supplementary materials**

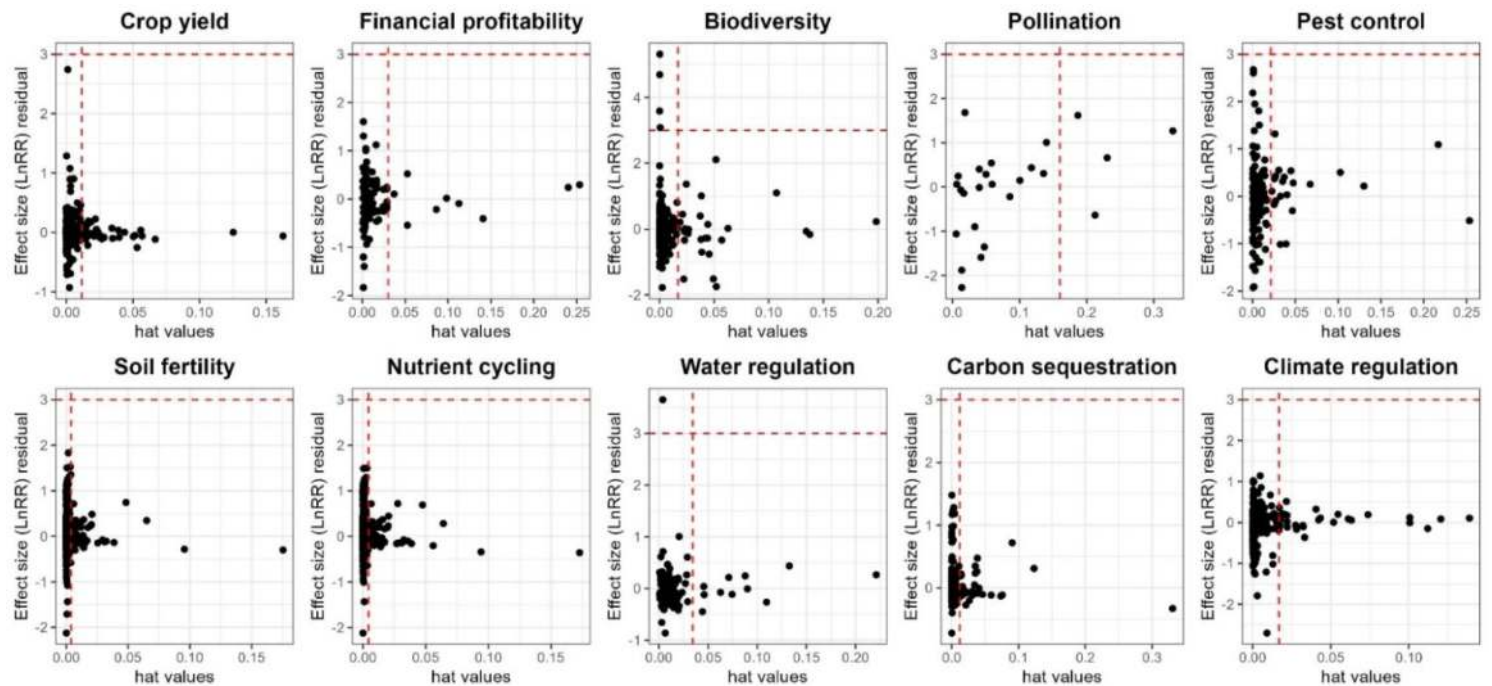
PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)



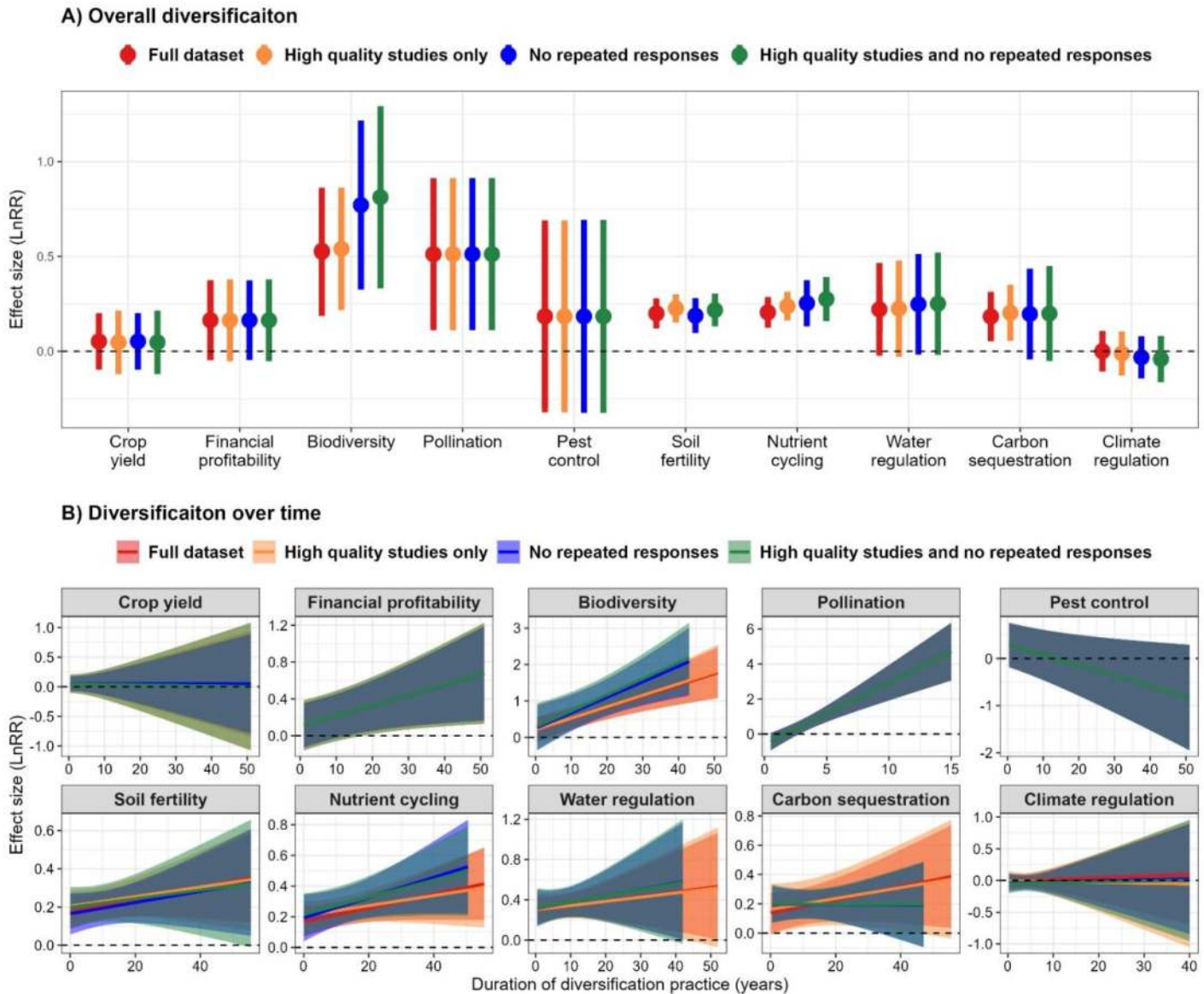
Supplementary Figure 1. PRISMA flow chart detailing the process of screening and extracting data from the studies obtained from the literature search.



Supplementary Figure 2. Publication bias based on funnel plots: Effect size (LnRR) and the precision or inverse standard error (1/SE) for each category of the effect sizes.



Supplementary Figure 3. Publication bias based on the influence of outliers: effect size $r(\text{LnRR})$ residual and the hat value for each category of the effect sizes.



Supplementary Figure 4. Sensitivity analysis for the second order meta-analysis: comparison of model results from the effect of diversification over time using the i) full dataset to the model results, ii) full dataset without repeated responses, iii) dataset with high quality studies only and iv) dataset with high quality studies only and no repeated studies. Some variables only show the regression slopes for high quality studies and no repeated responses (green): financial profitability - no repeated and only few low-quality studies; pollination - no repeated responses and all high-quality studies; pest control - few repeated responses and all high quality.

Supplementary Table 1. Research strings to extract studies from Web of science and Scopus.

Agricultural diversification systems	crop* OR "noncrop" OR intercropping OR "inter-cropping" OR "inter cropping" OR compost* OR till* OR "vegetation strip*" OR agroforest* OR inoculat* OR landscape OR organic OR fallow OR conventional OR fert* OR reduce* OR rotation OR "catch crop" OR amend*
Ecosystem services, yield, and financial profitability	divers* OR soil* OR biomass* OR water* OR pollutant* OR sediment* OR fodder OR emission* OR carbon* OR climat* OR pest* OR "biocontrol" OR weed* OR pollinat* OR fert* OR energ* OR resistance OR productiv* OR yield* OR "economic*" OR "profitability" OR "profit*" OR "econ*" OR "financ*" OR "gross returns" OR "gross income" OR "gross margin" OR "input*" OR "net return" OR "net income"
Type of studies	"meta-analysis" OR "metaanalysis" OR "meta analysis"
Publication date	Until 2022-12-31

Supplementary Table 2. Response variables and the categories to quantify the effects of diversification practices on socioeconomic, biological community, soil quality, and climate change mitigation factors adapted from Sánchez et al., 2022; Tamburini et al., 2020

	Variables' categories	Included response variables
Socioeconomic factors	Crop yield	Crop yield; Coffee yield; Productivity; Root biomass; Shoot biomass; Biomass; Main yield; Belowground biomass; Aboveground productivity; Above ground biomass; Yield loss; Total biomass; Fruit productivity; Timber productivity; System yield
	Financial profitability	Product price; Gross margins; Equipment costs; Labor costs; Herbicide costs; Management costs; Net return; Total cost; Gross return; Benefit/cost ratio; Gross income; Production costs; System revenue; Net present value; Price premiums; Gross revenue; Net revenue
Biological community	Biodiversity	Microbes richness; Actinomyces abundance; Animal species richness; Arthropods evenness; Arthropods richness; Autotrophs diversity; Bacteria abundance; Bacterial community diversity; Bacterial richness; Bird abundance; Bird and mammal diversity; Bird diversity; Bird richness; Community structure; Decomposers diversity; Decomposers richness; Detritivore richness; Earthworm abundance; Earthworm density; Earthworm richness; Forest species richness; Frugivores abundance; Frugivores diversity; Frugivores richness; Functional diversity; Fungal community diversity; Fungi abundance; Fungi richness; Fungi taxonomic diversity; Granivores diversity; Insectivores diversity; Insects abundance; Insects richness; Microbial diversity; Mite density; Natural enemy diversity; Nematode density; Nematode richness; Omnivore richness; Omnivores diversity; Phylogenetic diversity; Plant abundance; Plant species

		richness; Pollinator diversity; Pollinator richness; Predator richness; Prokaryotes taxonomic diversity; Protozoa richness; Richness AMF; Richness index; Shannon index; Species diversity; Species richness; Species richness and abundance; Springtail density
	Pollination	Pollinator richness; Pollinator diversity; Pollinator abundance; Pollinator evenness; Bee pollination
	Pest control	Arthropods pest abundance; Biocontrol; Disease damage; Herbivore abundance; Herbivore diversity; Herbivore evenness; Herbivore richness; Incidence of diseases; Incidence of pests; Insectivores diversity; Natural enemy diversity; Nematode damage; Omnivore richness; Omnivores diversity; Parasitoid abundance; Parasitoid richness; Pathogen control; Pathogen infestation; Pest damage; Pest infestation; Pests diversity; Plant damage; Predator abundance; Predator evenness; Predator richness; Proportion of healthy berries; Proportion of healthy leaves; Total natural enemy response; Total pest response; Weed abundance; Weed biomass; Weed control; Weed density; Weed infestation; Weed suppression; Weeds; Weeds diversity
Soil quality	Water regulation	Aggregate stability; Infiltration rate; N leaching; N leaching loss; N nitrate; N runoff; NO3 leaching; PAW - Plant Available Water; PWP - Permanent Wilting Point; Root length; Runoff; Sediment yield; Soil bulk density; Soil moisture; Soil porosity; Soil runoff; Soil saturated hydraulic conductivity; Soil water; Total P load; Total P to water; Volumetric water content; Water productivity; Water regulation; Water runoff; Water sediment; Water use; Water use efficiency

	Soil fertility	SOC; Abundance AMF; Actinomyces abundance; Aggregate stability; Alkali-hydrolysable nitrogen; Available K; Available N; Available P; Bacteria abundance; Bacteria biomass; Bacterial richness; Belowground carbon; Beta-glucosidase; Carbon sequestration; Catalase; Crop yield; Decomposers abundance; Decomposers diversity; Decomposers richness; Dehydrogenase; Detritivore abundance; Detritivore richness; Dissolved organic C; Earthworm abundance; Earthworm biomass; Earthworm density; Earthworm richness; Factor productivity of N; Fungi abundance; Fungi biomass; Fungi richness; Hydrolysable N; Invertase; Macroaggregates; Mean weight diameter of aggregates; Microaggregates; Microbes abundance; Microbes richness; Microbial biomass C; Microbial biomass N; Microbial diversity; Mite density; Mycorrhizal colonisation; N export; N leaching loss; N loss; N loss reduction; N nitrate; N use efficiency; Nematode abundance; Nematode density; Nematode richness; Nitrate N; Olsen P; Organic matter; P loss reduction; P use efficiency; PAW; Peroxidase; Phosphatase; Plant K uptake; Plant N uptake; Plant P uptake; Porosity; Potentially mineralizable N; PWP; Ratio MBC:MBN; Sediment yield; SHC; SOC stocks; Soil aggregation; Soil available K; Soil available P; Soil bulk density; Soil C; Soil C stocks; Soil C:N ratio; Soil C:P ratio; Soil catalase; Soil CEC; Soil EC; Soil enzyme activities; Soil fertility; Soil K; Soil loss; Soil moisture; Soil N; Soil N:P ratio; Soil N04; Soil organic matter; Soil P; Soil pH; Soil porosity; Soil runoff; Soil saturated hydraulic conductivity; Soil sucrose; Soil urease; Soil water-stable aggregation; Springtail density; Total C; Total C in system; Total K; Total N; Total N accumulation; Total P; Total SOC; Water sediment; Water stable aggregates
--	----------------	---

	Nutrient cycling	SOC; Actinomyces abundance; Alkali-hydrolysable nitrogen; Available K; Available N; Available P; Bacteria abundance; Bacteria biomass; Bacterial community diversity; Bacterial richness; Carbon sequestration; Colonization root length; Decomposition abundance; Decomposers diversity; Decomposers richness; Detritivore abundance; Detritivore evenness; Detritivore richness; Dissolved organic C; Dissolved organic N; Earthworm density; Earthworm abundance; Earthworm richness; Fungal community diversity; Fungi abundance; Fungi biomass; Fungi richness; Hydrolysable N; Microbes abundance; Microbes richness; Microbial biomass; Microbial biomass C; Microbial biomass N; Microbial diversity; Microbial Shannon index; Mite density; Mycorrhizal colonisation; N content; N export; N leaching; N leaching loss; N loss reduction; N nitrate; N runoff; N use efficiency; Nematode abundance; Nematode density; Nematode richness; Nitrate N; Organic matter; P concentration; P content; P loss reduction; Plant P uptake; Potentially mineralizable N; Ratio MBC:MBN; Root N concentration; Root nodules; Root P concentration; SOC stocks; Soil available K; Soil available P; Soil C; Soil C stocks; Soil K; Soil N; Soil organic matter; Soil P; Springtail density; Total C; Total C in system; Total K; Total N; Total N accumulation; Total P; Total SOC
Climate regulation	Carbon sequestration	SOC; Above ground carbon; Carbon sequestration; SOC stocks; Belowground carbon; Total C; Total SOC; Soil C stocks; Soil C; Dissolved organic C; Soil organic matter; Total C in system
	Climate regulation	CH ₄ emission; N use efficiency; CO ₂ emission; NO ₂ emission; Plant N uptake; N ₂ O emission; NH ₃ emission; Global warming potential; Biological N ₂ fixation; Factor productivity of N; Soil respiration; Total N accumulation; Soil NH ₄ ⁺ ; N ₂ fixation; N loss

Supplementary Table 3. Details of diversification practices under each category of agricultural diversification, similar to Tamburini et al., 2020

Practices categories	Practice details
Non-crop diversification	Landscape complexity; Mixed non crop species; Biofuel feed-stock; Integrating trees; Hedgerow; Afforestation; Buffer strips; Grass cover; Embedded natural; Sod cultivation; Animal-crop symbiosis; Fallow practices; Living mulch; Livestock incorporation; Agroforestry
Organic amendment	Biosolids application; Residue retention; Manure application; Straw amendment; Organic amendment; Biochar amendment; Compost fertilizer; Organic fertilizer

Crop diversification	Crop rotation; Agroforestry; Cover cropping; Strip cropping; Intercropping; Cultivar mixture; Polyculture; Catch cropping; Mixed cropping
Reduced tillage	Reduced tillage; No tillage; Direct seeding; Subsoiling tillage; Minimum tillage
Organic farming	Organic farming; Organic management
Inoculation	AMF inoculation; EMF inoculation; Rhizobia co-inoculation

Supplementary Table 4. List of information for assessing the quality of each meta-analysis study included in this study. Each information was scored one when absent and two when present. The score of each study was equal to the total scores of all information.

	Quality assessment criteria	Score
Methods	Literature search method provided	No=1 , Yes=2 Total score=16 Quality Assessment 8-13 >> Low quality 14-16 >> High quality
	Inclusion/exclusion criteria of original studies reported	
	Definition of the control group included	
	Heterogeneity assessment included	
	Weighting procedure reported	
Results	Number of original studies reported	
	Effect size average and CIs included	
	Sensitivity analysis control included	

Supplementary Table 5. Results of Egger's regression test to detect asymmetry of funnel plots for each category of the response variables.

		estimate	se	zval	pval	ci.lb	ci.ub	Sign.
Crop yield	intrcpt	0.043	0.0812	0.5297	0.5963	-0.1162	0.2022	
	mods	0.2658	0.6918	0.3842	0.7008	-1.0901	1.6217	
Finance profitability	intrcpt	0.1157	0.1629	0.7103	0.4775	-0.2035	0.4349	
	mods	0.157	0.3293	0.4769	0.6335	-0.4883	0.8023	
Biodiversity	intrcpt	0.5037	0.2122	2.3742	0.0176	0.0879	0.9196	*
	mods	0.15	0.6157	0.2436	0.8075	-1.0568	1.3568	
Pollination	intrcpt	-0.1903	0.246	-0.7736	0.4392	-0.6724	0.2918	
	mods	1.5605	0.5931	2.6312	0.0085	0.3981	2.723	**
Pest control	intrcpt	0.2979	0.2731	1.0905	0.2755	-0.2375	0.8332	
	mods	-0.3606	0.4025	-0.8958	0.3703	-1.1495	0.4283	
Water regulation	intrcpt	0.1264	0.075	1.6858	0.0918	-0.0205	0.2733	.
	mods	1.0992	0.43	2.5562	0.0106	0.2564	1.942	*
Soil fertility	intrcpt	0.1647	0.0404	4.0742	<.0001	0.0855	0.2439	***

	mods	0.541	0.2586	2.0924	0.0364	0.0342	1.0479	*
Nutrient cycling	intrcpt	0.175	0.0455	3.8479	0.0001	0.0859	0.2642	***
	mods	0.4493	0.2736	1.6424	0.1005	-0.0869	0.9855	
Carbon sequestration	intrcpt	0.1422	0.0728	1.9529	0.0508	-0.0005	0.2848	.
	mods	0.8835	0.4404	2.0061	0.0448	0.0203	1.7467	*
Climate regulation	intrcpt	0.0594	0.0674	0.8808	0.3784	-0.0728	0.1915	
	mods	-0.7442	0.459	-1.6216	0.1049	-1.6438	0.1553	

Supplementary Table 6. Results of fail-safe N analysis based on Rosenthal method for each category of the response variables.

	Fail-safe N	Observed significant level
Crop yield	382509	<.0001
Financial profitability	1259	<.0001
Biodiversity	713221	<.0001
Pollination	348	<.0001
Water regulation	234372	<.0001
Pest control	1483	<.0001
Soil fertility	15021430	<.0001
Nutrient cycling	11475248	<.0001
Carbon sequestration	2709693	<.0001
Climate regulation	8611	<.0001

Supplementary Table 7. Results of publication bias test based on Likelihood Ratio Test for each category of the response variables.

	X^2	p-value
Crop yield	0.1652429	0.68437
Financial profitability	1.992073	0.15812
Biodiversity	2.227789	0.39767
Pollination	0.1792251	0.67204
Pest control	0.7644067	0.38195
Water regulation	0.07537393	0.78367
Soil fertility	1.232491	0.26692
Nutrient cycling	3.145582	0.076133
Carbon sequestration	2.71357	0.099498
Climate regulation	2.775196	0.28873